

Multi Layered Anomaly Detection Framework for Network Security using Machine Learning

A Project Report

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Technology in Computer Science and Engineering (Data Science)

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LUCKNOW**

May, 2026

Declaration

We hereby declare that this project report entitled ‘**Multi Layered Anomaly Detection Framework for Network Security using Machine Learning**’ submitted to the Department of Computer Science and Engineering, Ajay Kumar Garg Engineering College, Ghaziabad in partial fulfillment of the requirements for the degree of *Bachelor of Technology* is an authentic record of our original work carried out during the academic year 2022–2026.

The work contained in this report has not been submitted in part or in full to any other university or institution for the award of any degree or diploma. All sources of information used in this report have been duly acknowledged and referenced.

We have self-checked this report for plagiarism using appropriate tools and confirm that it meets the prescribed academic integrity standards.

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Certificate

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Abstract

With the increasing complexity of cyber attacks, traditional single-layer security mechanisms are not sufficient to detect complex anomalies. The current project aims to design a multi-layer anomaly detection system for network security using machine learning techniques at three different hierarchical levels: packet-level, flow-level, and behaviour-level. The packet-level module uses machine learning algorithms like random forest to detect unusual patterns in packet headers. The flow-level module uses autoencoders to detect unusual patterns in flow-level features like flow duration, packet counts, etc. The behaviour-level module uses recurrent neural networks like GRU to learn normal patterns of user or system behaviour over time to detect unusual patterns. The fusion module combines the results of all the above modules to detect anomalies.

The system was implemented as a real-time simulation system with a user-friendly dashboard to view the live network traffic. The system was also implemented on benchmark intrusion datasets like CIC-IDS2017, NSL-KDD, etc., with rigorous preprocessing of the data. The system was also tested for metrics like accuracy, false positive rate, recall, etc. The results showed that the system performed better for multi-layer anomaly detection compared to single-layer anomaly detection. For example, the system was found to have 97% accuracy for packet-level and flow-level anomaly detection. The fusion module was found to have an accuracy of about 96% with a false positive rate of less than 6%. The system was also found to have good generalisation performance for different intrusion detection datasets. The system was found to have an accuracy of about 95% for cross-dataset evaluation. The system was found to have high precision, recall, and F1-score for all the different scenarios of intrusion detection.

In conclusion, the system was found to perform better for multi-layer anomaly detection compared to single-layer anomaly detection. The system was found to have high accuracy with a low false positive rate for all scenarios of intrusion detection. The system was found to be practical for implementation based on the results obtained for the simulated system.

Keywords: Machine Learning, Network Security, Cyber Attacks, Autoencoders, Random Forest

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Chapter 1

Introduction

1.1 Background and Motivation

In the modern digital era, the rapid advancement of information technology has led to an unprecedented dependence on computer networks and internet-based systems across almost every sector, including finance, healthcare, education, government, and industrial operations. Organizations today rely heavily on continuous data exchange, cloud computing, IoT devices, and distributed systems to maintain efficiency and scalability. However, this increased reliance has simultaneously exposed network infrastructures to a wide range of sophisticated cyber threats [1, 2].

Cybersecurity has become a critical concern due to the exponential rise in cyberattacks such as Distributed Denial of Service (DDoS), phishing, ransomware, botnets, insider attacks, and zero-day vulnerabilities [3]. Traditional security mechanisms such as firewalls, antivirus software, and signature-based intrusion detection systems (IDS) are no longer sufficient to combat modern threats [4]. These systems primarily rely on predefined rules or known attack signatures, making them ineffective against unknown or evolving attack patterns.

A major limitation of conventional approaches is their inability to detect anomalies that do not match previously known patterns [5]. Attackers today employ advanced techniques such as polymorphic malware, encrypted communication channels, and stealth-based intrusion strategies that can easily bypass traditional defenses. Additionally, insider threats—where legitimate users misuse their access—pose a significant challenge, as they often appear normal at the surface level.

To address these limitations, there is a growing need for intelligent and adaptive security systems capable of identifying unusual patterns in network behavior. Anomaly detection has emerged as a promising solution [5, 6], as it focuses on identifying deviations from normal behavior rather than relying solely on known attack signatures. However, single-layer anomaly detection systems still fall short because they analyze only one aspect of network activity.

This project is motivated by the need to develop a multi-layered anomaly detection framework that analyzes network traffic from multiple perspectives [7]—packet-level, flow-level, and behavioral-level—thereby improving detection accuracy and reducing false positives. By integrating machine learning techniques into this layered approach, the system aims to provide a more robust, scalable, and intelligent solution for modern network security chal-

lenges.

1.2 Problem Statement

Despite the availability of numerous intrusion detection systems, accurately identifying malicious activities in network traffic remains a challenging problem [1]. Existing systems often focus on a single dimension of network analysis, such as packet inspection or traffic flow monitoring. This narrow focus limits their ability to detect complex, multi-vector attacks that span across different layers of the network.

One of the primary challenges in anomaly detection is distinguishing between legitimate variations in network behavior and actual malicious activities [8]. For instance, a sudden spike in network traffic may be due to normal user activity or an ongoing attack such as a DDoS. Single-layer systems often misinterpret such variations, resulting in high false positive rates. Similarly, some attacks are designed to mimic normal behavior, making them difficult to detect without a deeper, multi-dimensional analysis.

Another significant issue is the inability of traditional systems to adapt to changing network environments [4]. Network behavior evolves over time due to changes in user patterns, system upgrades, and application usage. Static rule-based systems fail to adapt to these changes, leading to decreased performance over time.

Furthermore, modern cyberattacks are increasingly sophisticated and often involve multiple stages, including reconnaissance, exploitation, lateral movement, and data exfiltration. Detecting such attacks requires analyzing network activity at multiple levels simultaneously. However, most existing systems lack integration across different layers of analysis, resulting in fragmented detection mechanisms [2].

1.3 Research Objectives

This research is all about designing a layered anomaly detection system for network security, which improves the detection of cyber threats more accurately and with higher reliability. Instead of a single layer for detection, the system analyzes network traffic at three different layers: the packet layer, the flow layer, and the behavior layer.

We are also going to select and apply machine learning techniques specific to the layers for which the techniques are suitable. For the packet layer, the system uses the Random Forest technique to analyze the features of the network traffic [9], including the type of protocol, port numbers, and size of the packets, to identify any suspicious or erroneous packets. For the flow layer, the system makes use of the autoencoder technique [10, 11], which analyzes the normal network traffic features, including the duration of the flows, the

number of packets, and the amount of data transfer, making it simpler to identify any unusual flows. For the behavior layer, the system makes use of the GRU neural network technique, which analyzes the temporal behavior of users and systems to identify any insider threats or changing patterns of cyber attacks.

1.4 Novelty and Originality

The project is a step forward from the previous one-note security approach [4] that focuses on the use of traffic signatures or a single detection layer. The project offers a multi-layer anomaly detection model that is designed to enhance the reliability and accuracy with which threats are detected. The anomalies are detected separately in the three layers: packet level, flow level, and behavior level. This way, the system is able to look at the anomalies from different angles. The different layers of the model focus on different aspects of the network. This way, the system is able to look at the anomalies from different angles. The results from all the three levels are cross-checked to enhance the accuracy of the detection while reducing false positives [7, 11]. The approach combines the use of heuristic analysis with the best machine learning techniques. This way, the system is able to look deeper into the anomalies. The approach combines different viewpoints of the analysis. This way, the system is able to detect anomalies that may otherwise go undetected. The framework also offers a simulations-based approach to show how the system works without the need for any special hardware. This way, the system is able to show its flexibility and scalability. This makes the approach suitable for use in different situations. The approach is a practical way to enhance the current cybersecurity standards.

1.5 Scope of the Project

The primary focus of this project is on anomaly detection in network traffic using a simulated real-time environment. Instead of deploying the system directly within a live network infrastructure, the study makes use of well-established benchmark datasets that closely replicate real-world network behavior [12, 13, 14]. These datasets contain a mixture of normal and malicious traffic patterns, enabling the system to operate in conditions that resemble practical scenarios without the risks associated with live deployment.

By working within this simulated environment, the system is able to analyse a wide variety of network activities and identify unusual patterns that may indicate potential cyber threats. This includes detecting deviations from normal behaviour, unusual traffic spikes, and patterns commonly associated with different types of attacks. The use of simulated data also allows for controlled experimentation, where different attack scenarios and traffic conditions can be introduced [12] deliberately to evaluate how effectively the system responds.

Another important aspect of this work is the flexibility it provides in experimenting with different machine learning techniques [1]. Various models can be implemented, tested, and compared to understand their effectiveness in identifying anomalies. This enables a deeper analysis of model performance in terms of accuracy, detection rate, and reliability. The focus remains strictly on detection and analysis rather than prevention, distinguishing this system from traditional intrusion prevention systems. The goal is to build a strong detection mechanism that can later be integrated into larger security frameworks.

The simulated setup also makes the system highly adaptable and cost-effective. Since it does not require deployment in a real-time production network, it eliminates risks such as unintended disruptions or security breaches during testing. At the same time, it provides a scalable foundation that can be extended in the future. The architecture designed in this project can be further developed for use in real-world environments such as enterprise networks, cloud-based systems, or large-scale distributed infrastructures.

Additionally, the controlled environment allows for efficient performance evaluation and fine-tuning of the system. It becomes easier to measure how the system behaves under different conditions and to improve its accuracy over time. The framework also creates opportunities for future enhancements, such as the integration of real-time alert mechanisms, automated response systems, and advanced visualization tools.

Overall, the project establishes a strong base for developing a robust and flexible anomaly detection system. While the current implementation focuses on simulated data, the design and approach ensure that it can be scaled and adapted for practical, real-world applications in the field of network security.

1.6 Report Organisation

This report is organised as follows.

Chapter 1 introduces the background and motivation of the study, highlighting the growing importance of robust cybersecurity mechanisms in modern digital infrastructures. It outlines the research objectives, defines the problem statement associated with the limitations of existing intrusion detection systems, and presents the scope and overall structure of the report.

Chapter 2 presents a comprehensive review of existing literature related to network security, intrusion detection systems, and anomaly detection techniques. It covers traditional approaches such as signature-based and anomaly-based detection methods, along with recent advancements in machine learning and deep learning. The chapter critically analyses existing research contributions and identifies the limitations of single-layered detection systems, thereby establishing the need for a more efficient multi-layered framework.

Chapter 3 describes the research methodology adopted in this study, including data col-

lection, preprocessing, and feature extraction. It further elaborates on the design and development of the proposed multi-layer anomaly detection framework and explains how different detection layers are structured and integrated. The evaluation metrics and experimental setup used for performance analysis are also discussed.

Chapter 4 explains the system design, architecture, and implementation details of the proposed framework. It provides an in-depth description of system components such as the attacker simulation panel, backend processing unit, and the Flutter-based dashboard. The workflow, data flow, and integration of modules are also elaborated.

Chapter 5 presents the experimental results obtained from the implementation of the proposed system and provides a detailed analysis of its performance. It evaluates the effectiveness of the multi-layer approach in detecting anomalies compared to traditional methods, highlighting improvements in accuracy, detection rate, and reliability.

Chapter 6 discusses the societal impact of the proposed system and its alignment with Sustainable Development Goals (SDGs). It emphasizes the role of secure digital systems in sectors such as healthcare, finance, education, and governance, and highlights how enhanced cybersecurity contributes to building resilient and sustainable technological ecosystems.

Chapter 7 addresses Universal Human Values, focusing on ethical considerations, trust, and the importance of developing secure and human-centric digital environments. It reflects on the role of cybersecurity in ensuring responsible use of technology and maintaining public confidence in digital systems.

Chapter 8 concludes the report by summarising the key findings, contributions, and limitations of the study, and outlines potential directions for future research and further enhancements to the proposed multi-layer anomaly detection framework.

Chapter 2

Literature Review

2.1 Historical Overview

Evolution of Network Security and Anomaly Detection

Network security has continued to evolve with the increasingly complex digital landscape. Traditionally, network security was focused on perimeter defenses, which included firewalls and signature-based intrusion detection systems [4]. The latter used signature-based detection, which compared network traffic against known signature patterns of malicious attacks. This was moderately successful against known threats, but not so good against unknown threats. This was because attackers were becoming increasingly sophisticated, leading to the emergence of zero-day attacks, which were not detected by signature-based intrusion detection systems.

In the late 1990s, network security was evolving, with anomaly detection emerging as a new approach to network security [5]. Unlike signature-based systems, which relied on known malicious patterns, anomaly detection focused on identifying unusual patterns of network traffic. Initially, statistical methods were used for anomaly detection [5], which later involved rule-based systems. Although anomaly detection was moderately successful, the rise of zero-day attacks meant that signature-based systems were not as good as anomaly detection systems. However, the latter had the disadvantage of higher false alarms, which were often mistaken for malicious activity.

Recently, the rise of cloud computing, big data, and high-speed networks has continued to transform the security landscape. Today, intrusion detection systems rely on sophisticated machine learning techniques [1, 6], which are used for efficient intrusion detection. The smart systems are used to identify complex patterns of network traffic, making them highly efficient for the detection of complex cyber threats.

2.2 Recent and State-of-the-Art Work

With the increasing complexity of cyber threats, traditional intrusion detection systems have been enhanced using machine learning and deep learning techniques. Modern approaches focus on adaptability, scalability, and the ability to detect unknown or zero-day attacks.

2.2.1 Machine Learning-Based Intrusion Detection

Machine learning techniques such as Decision Trees, Random Forests, Support Vector Machines (SVM), and k-Nearest Neighbors (k-NN) are widely used for classifying network traffic [1, 8]. Among these, Random Forest is particularly effective due to its robustness and ability to handle high-dimensional data [9]. SVM models also provide strong classification performance in complex scenarios.

However, supervised models rely on labeled datasets and often fail to detect unseen or zero-day attacks, limiting their effectiveness in dynamic environments.

2.2.2 Unsupervised and Semi-Supervised Approaches

To reduce dependence on labeled data, unsupervised methods such as k-Means, DBSCAN, and Isolation Forest are used to identify anomalies based on deviations from normal behavior. Isolation Forest, in particular, is effective in detecting rare and unknown attacks [9].

Semi-supervised learning combines labeled and unlabeled data to improve detection performance. Despite their advantages, these approaches may result in higher false positive rates.

2.2.3 Deep Learning-Based Detection

Deep learning techniques have significantly improved anomaly detection capabilities [15, 1]. Autoencoders are widely used to detect anomalies based on reconstruction error [10, 11], while Recurrent Neural Networks (RNN), including LSTM and GRU, capture temporal patterns in network behavior.

Convolutional Neural Networks (CNN) have also been adapted for analyzing structured network data. Although deep learning models provide high accuracy, they require large datasets and high computational resources.

2.2.4 Hybrid and Real-Time Detection Systems

Recent research focuses on hybrid and ensemble approaches that combine multiple algorithms [7, 11] to improve accuracy and reduce false positives. These systems often integrate machine learning and deep learning models across different layers of analysis.

Additionally, real-time detection systems using big data technologies have been developed to process large volumes of network traffic efficiently. However, maintaining both accuracy and low latency remains a challenge.

2.2.5 Limitations and Research Direction

Despite advancements, existing systems still face challenges such as high false positive rates, difficulty in detecting stealthy attacks, and lack of multi-layer integration. These limitations highlight the need for multi-layered anomaly detection frameworks that analyze network traffic at packet, flow, and behavioral levels for improved accuracy and reliability.

2.3 Identification of Research Gap

Despite significant advancements in network anomaly detection, existing approaches still exhibit several limitations that reduce their effectiveness in real-world environments. These gaps highlight the need for more comprehensive and adaptive solutions.

2.3.1 Limitations of Single-Layer Detection

Most traditional intrusion detection systems focus on a single layer of network analysis [4], such as packet-level inspection or flow-level monitoring. While these approaches can detect certain types of attacks, they fail to provide a complete understanding of network behavior. Sophisticated attacks often span multiple layers and may appear normal when analyzed from only one perspective, leading to missed detections.

2.3.2 Lack of Integrated Analysis

Many existing systems analyze different aspects of network traffic independently without integrating their outputs. This lack of coordination creates blind spots [2], where anomalies that are subtle at one level but significant at another remain undetected. The absence of a unified decision-making mechanism limits overall detection accuracy.

2.3.3 High False Positive and False Negative Rates

Anomaly detection systems often struggle to distinguish between legitimate variations in network activity and actual malicious behavior. This results in high false positive rates [8], which can overwhelm system administrators, as well as false negatives, where real threats go unnoticed. Maintaining a balance between sensitivity and accuracy remains a major challenge.

2.3.4 Inability to Detect Advanced and Evolving Threats

Modern cyberattacks are highly sophisticated and continuously evolving. Techniques such as polymorphic malware, insider threats, and stealth attacks are designed to mimic normal

behavior [3] and evade detection. Traditional and even some machine learning-based systems fail to adapt quickly to these changing patterns, reducing their effectiveness against zero-day attacks.

2.3.5 Scalability and Real-Time Processing Challenges

With the rapid growth of network traffic, existing systems often face challenges in handling large-scale data efficiently. Real-time detection requires high computational power and optimized algorithms [16]. Many current approaches struggle to maintain performance and accuracy in high-speed or large-scale network environments.

2.3.6 Research Direction

The limitations identified above indicate the need for a more holistic and adaptive approach to anomaly detection. A multi-layered framework that integrates packet-level, flow-level, and behavioral-level analysis [7] can provide a more comprehensive view of network activity. By combining multiple machine learning techniques [1] and employing a fusion-based decision mechanism, it is possible to improve detection accuracy, reduce false alarms, and enhance the system's ability to detect complex and evolving cyber threats.

Chapter 3

Research Methodology

3.1 Research Approach

This research proposes a multi-layered anomaly detection framework designed to identify malicious network activities, particularly Distributed Denial of Service (DDoS) attacks. With the increasing complexity of cyber threats, traditional intrusion detection systems are no longer sufficient, as they rely on static rules and single-layer analysis. The proposed system addresses these limitations by analyzing network traffic at multiple levels, thereby improving detection accuracy and robustness.

The methodology follows a data-driven experimental approach in which both real-world datasets and simulated traffic are used. Publicly available datasets such as CICIDS2017, NSL-KDD, and UNSW-NB15 provide labeled data for training and testing the models. In addition, a custom attacker panel is used to generate synthetic traffic, enabling real-time simulation of attack scenarios.

The system is designed to process data through multiple stages, including data collection, preprocessing, feature extraction, multi-layer detection, and result visualization. Each stage is carefully structured to ensure efficient processing and accurate detection.

3.2 Proposed System

The proposed framework consists of multiple interconnected modules that work together to detect anomalies in network traffic. The modular design ensures scalability and flexibility, allowing the system to adapt to different network environments.

3.2.1 System Modules Overview

The proposed anomaly detection framework is designed as a modular system in which each component performs a specific function within the overall detection pipeline. This modular architecture ensures scalability, flexibility, and ease of maintenance, allowing individual modules to be improved or replaced without affecting the entire system.

The system processes network traffic in a sequential manner, beginning with data collection and ending with visualization of detection results. Each module is interconnected and contributes to the overall goal of accurately identifying anomalies in network traffic.

The key modules of the system are described below:

Table 3.1: Overview of System Modules

Module	Description
Data Collection Module	Responsible for gathering network traffic data from multiple sources including benchmark datasets and simulated traffic generated through the attacker panel. It ensures that both normal and malicious traffic patterns are available for analysis.
Data Preprocessing Module	Handles cleaning and transformation of raw data. It removes inconsistencies, handles missing values, normalizes numerical features, and encodes categorical variables to prepare the data for machine learning models.
Feature Extraction Module	Extracts relevant features from network traffic at packet-level, flow-level, and behavioral-level. These features represent the characteristics of network activity and are used as input for detection models.
Multi-Layer Detection Module	The core component of the system that applies different machine learning models at each layer. It analyzes data from multiple perspectives to detect anomalies and improve detection accuracy.
Decision Fusion Module	Combines the outputs of all detection layers using a majority voting mechanism. It ensures that the final decision is reliable and not dependent on a single model.
Monitoring and Visualization Module	Provides real-time visualization of network activity and detected anomalies through the Flutter dashboard. It also integrates with the attacker panel for simulation and testing.

Each of these modules plays a critical role in the overall functioning of the system. The modular design allows the system to handle large-scale network traffic efficiently while maintaining high detection accuracy. Furthermore, this structure supports future enhancements, such as integration with real-time network monitoring systems or deployment in cloud-based environments.

The interaction between these modules ensures a smooth flow of data from input to output. Data collected from various sources is first preprocessed and transformed into meaningful features. These features are then analyzed by multiple detection models, and the results are combined to produce the final decision. The output is displayed on the dashboard, enabling users to monitor network activity and respond to potential threats effectively.

3.2.2 Data Collection Module

The data collection module is responsible for gathering network traffic data from multiple sources. The primary datasets used in this research include CICIDS2017, NSL-KDD, and UNSW-NB15. These datasets contain a wide range of attack scenarios and normal traffic patterns, making them suitable for training machine learning models.

In addition to these datasets, a custom attacker panel is developed to simulate real-time attacks. This panel generates synthetic traffic patterns such as high-volume requests, irregular packet bursts, and repeated connections. This allows the system to be evaluated under realistic conditions.

Key characteristics of the collected data include:

- Presence of both normal and malicious traffic
- Diversity in attack types
- High dimensional feature space

3.2.3 Data Preprocessing Module

The preprocessing stage ensures that the data is clean, consistent, and suitable for model training. Raw network data often contains noise, missing values, and redundant features, which can negatively impact model performance.

The preprocessing pipeline includes the following steps:

- Removal of duplicate and irrelevant records
- Handling missing values using statistical methods
- Normalization of numerical features
- Encoding of categorical variables
- Feature selection to reduce dimensionality

These steps help improve the quality of the dataset and enhance model efficiency.

3.2.4 Feature Extraction Module

Feature extraction is a crucial step that transforms raw data into meaningful inputs for the detection models. Features are extracted at three levels to capture different aspects of network behavior.

Packet-Level Features:

Table 3.2: Preprocessing Techniques

Technique	Purpose
Data Cleaning	Remove noise and duplicates
Normalization	Scale features uniformly
Encoding	Convert categorical data
Feature Selection	Reduce redundancy

- Protocol type
- Packet size
- Source and destination IP
- TCP flags

Flow-Level Features:

- Flow duration
- Number of packets
- Total bytes transferred
- Inter-arrival time

Behavioral-Level Features:

- Connection frequency
- Session duration
- Activity trends over time

Table 3.3: Feature Categories

Level	Features
Packet-Level	Protocol, packet size, flags
Flow-Level	Duration, bytes, packets
Behavioral-Level	Frequency, session patterns

3.2.5 Multi-Layer Detection Module

The detection module is the core of the system and consists of three layers, each using a different machine learning approach.

- **Packet-Level Detection:** Uses Random Forest and Isolation Forest to detect anomalies in packets.
- **Flow-Level Detection:** Uses Autoencoder to identify abnormal communication patterns.
- **Behavioral-Level Detection:** Uses GRU to analyze sequential data.

Each model is trained independently and contributes to the final decision.

Table 3.4: Detection Models

Layer	Model	Purpose
Packet-Level	Random Forest	Packet anomaly detection
Flow-Level	Autoencoder	Pattern deviation detection
Behavioral-Level	GRU	Sequential anomaly detection

3.2.6 Decision Fusion Module

To improve detection reliability, the outputs of all three layers are combined using a majority voting mechanism. This ensures that the final decision is based on multiple perspectives rather than a single model.

$$D_{final} = mode(P_p, P_f, P_b)$$

This approach reduces false positives and increases overall accuracy.

3.2.7 Monitoring and Visualization Module

The monitoring and visualization module acts as the final layer of the proposed anomaly detection system, providing real-time insights into network activity and detected anomalies. This module is responsible for presenting the processed results from the multi-layer detection engine in a structured and user-friendly manner, enabling administrators to quickly interpret system outputs and respond to potential threats.

The system integrates outputs from packet-level, flow-level, and behavioural-level analysis, which are combined using a fusion mechanism. These results are then visualized through an interactive interface that supports real-time monitoring and decision-making.

Attacker Panel:

The attacker panel is designed to simulate network traffic and generate various attack scenarios. It plays a critical role in testing the robustness and effectiveness of the anomaly detection system under controlled conditions.

- Simulates different types of attacks such as abnormal traffic bursts and repeated requests
- Generates real-time network traffic using packet capture tools like **tshark**
- Streams captured packets to the backend for processing and analysis
- Enables controlled testing of system performance under different attack scenarios

Flutter Dashboard:

The Flutter-based dashboard serves as the primary interface for monitoring network activity and viewing anomaly detection results. It is connected to the backend system, which processes incoming data and sends results to the dashboard in real time.

- Displays real-time network traffic data and statistics
- Shows detected anomalies and alert notifications
- Provides graphical insights such as trends and patterns in traffic
- Presents evaluation metrics including Accuracy, Precision, Recall, and F1-score

The dashboard is designed with a clean and intuitive layout, ensuring ease of use and efficient navigation between different system components. This allows administrators to quickly identify suspicious activities and take necessary actions.

System Flow Visualization

Figure 3.1 illustrates the overall working of the monitoring and visualization module. It shows how incoming network traffic is processed across multiple detection layers and how the final results are presented through the dashboard.

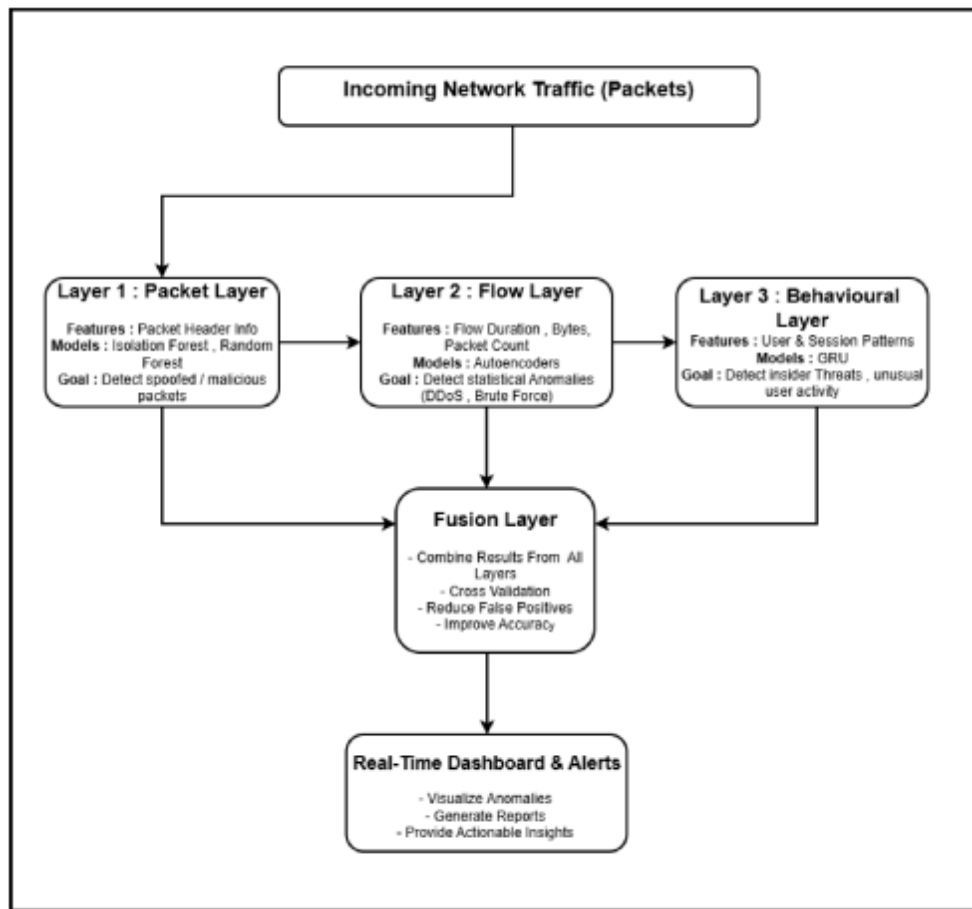


Figure 3.1: Multi-Layer Detection and Monitoring Flow

The diagram highlights the flow of data from incoming packets through packet-level, flow-level, and behavioural-level analysis. The outputs from these layers are combined in the fusion layer, and the final decision is visualized through the dashboard, enabling real-time monitoring and effective threat detection.

3.3 Experimental Setup

The system is evaluated using a controlled experimental setup.

- Training Data: 70%
- Testing Data: 30%
- Cross-validation: 5-fold

This setup ensures fair evaluation and avoids overfitting.

3.4 Tools and Technologies

Table 3.5: Tools and Technologies Used

Category	Technology	Purpose
Programming	Python	ML implementation
ML Libraries	TensorFlow, Scikit-learn	Model training
Frontend	Flutter	Dashboard UI
Simulation	Attacker Panel	Traffic generation
Data Processing	Pandas, NumPy	Preprocessing
Visualization	Matplotlib	Graphs

3.5 Evaluation Metrics

The system performance is evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- False Positive Rate

These metrics provide a comprehensive evaluation of the detection system.

3.6 Summary

The proposed methodology provides a detailed and structured approach for anomaly detection using multi-layer analysis. By integrating machine learning models, real-time simulation, and visualization tools, the system achieves improved accuracy, scalability, and usability in modern network environments.

Chapter 4

Implementation

4.1 System Architecture Overview

The implemented system is based on a multi-layered architecture for anomaly detection that leverages machine learning techniques to identify malicious activities within network traffic. The architecture is designed to overcome the limitations of traditional intrusion detection systems by incorporating multiple analytical perspectives. Instead of relying on a single detection mechanism, the system analyzes network data at three distinct levels, namely packet-level analysis, flow-level analysis, and behavioural-level analysis. Each of these layers focuses on a specific aspect of network traffic, enabling the system to detect both simple and complex attack patterns more effectively.

The three analytical layers operate independently on the extracted features of the network data. Packet-level analysis examines individual packets and their attributes, such as protocol type, packet size, and header information. Flow-level analysis focuses on communication patterns between network entities, including metrics such as flow duration, number of packets, and data transfer volume. Behavioural-level analysis captures long-term patterns and trends in network activity, such as connection frequency and session behaviour. By combining insights from all three layers, the system achieves a more comprehensive understanding of network behaviour and improves detection accuracy.

The system begins with the **Dataset Ingestion and Simulation Layer**, which acts as the entry point for all incoming data. This layer is responsible for collecting network traffic data from publicly available benchmark datasets such as CICIDS2017, NSL-KDD, and UNSW-NB15. These datasets provide a diverse range of normal and malicious traffic scenarios, including modern attack types. In addition to static datasets, this layer also incorporates simulated traffic generated through the attacker panel. The simulation process streams data sequentially to mimic real-time network conditions, ensuring that the system can handle dynamic and continuous data input. The data is structured in the form of packets and flows, closely resembling actual network traffic.

Once the data is ingested, it is passed to the **Feature Extraction and Preprocessing Center**. This component plays a crucial role in preparing the raw data for analysis. The preprocessing stage involves multiple steps, including data cleaning, handling of missing or inconsistent values, normalization of numerical attributes, and encoding of categorical variables. These steps ensure that the data is consistent and suitable for machine learning models. Following preprocessing, relevant features are extracted from the data. These fea-

tures are categorized based on the three analytical levels, allowing each detection layer to operate on the most appropriate set of attributes.

After preprocessing and feature extraction, the data is forwarded to the **Anomaly Detection and Fusion Engine**, which forms the core of the system. This engine consists of three independent machine learning models, each corresponding to one of the analytical layers. The packet-level model focuses on detecting anomalies in individual packets using algorithms such as Random Forest and Isolation Forest. The flow-level model uses deep learning techniques, such as Autoencoders, to identify abnormal communication patterns within network flows. The behavioural-level model employs sequence-based models, such as GRU, to analyze temporal patterns and detect long-term anomalies in network activity.

Each model processes the input features independently and produces a prediction indicating whether the observed behaviour is normal or anomalous. These predictions are then passed to the fusion component, where they are combined using a fusion-based decision mechanism. The system uses a majority voting approach, where the final decision is determined based on the agreement among the three models. This approach enhances reliability by reducing dependence on any single model and minimizing false positives.

Finally, the system generates results through a **dashboard interface**, which serves as the visualization and monitoring layer. The dashboard, implemented using Flutter, provides a user-friendly interface for administrators to observe network activity in real time. It displays detected anomalies, traffic statistics, and graphical representations of network behaviour. This allows administrators to quickly identify potential threats, analyze attack patterns, and take appropriate actions. The integration of visualization with the detection system enhances usability and ensures that the system is practical for real-world deployment.

4.2 Module-wise Description

4.2.1 Module 1: Data Ingestion and Traffic Streaming

This module serves as the entry point of the entire anomaly detection system and is responsible for collecting, organizing, and streaming network traffic data into the processing pipeline. It plays a crucial role in ensuring that the system receives a continuous and structured flow of data for analysis.

The system utilizes publicly available intrusion detection datasets such as CICIDS2017, NSL-KDD, and UNSW-NB15. These datasets contain a wide range of normal and malicious traffic patterns, including modern cyberattacks. The diversity of these datasets ensures that the system is trained on realistic scenarios and can generalize effectively to unseen data.

In addition to static datasets, this module also integrates simulated traffic generated through the attacker panel. The attacker panel produces synthetic traffic patterns that mimic

real-world attack scenarios such as high-volume packet bursts, repeated connection attempts, and abnormal traffic spikes. This allows the system to operate under dynamic conditions and evaluate its performance in near real-time environments.

Primary Functions

- Loading dataset records from multiple sources
- Filtering relevant network traffic based on protocol or attack type
- Sequential streaming of packet and flow data
- Simulation of real-time traffic using attacker panel

Inputs

- Network traffic datasets
- Simulated traffic from attacker panel

Outputs

- Continuous stream of packet, flow, and session data

Key Logic

The module operates by sequentially feeding data records into the system in a time-ordered manner. This approach simulates real-time network conditions and ensures that the detection models process data dynamically rather than in batches. This significantly improves the practical applicability of the system.

Table 4.1: Data Ingestion Characteristics

Aspect	Description
Data Sources	CICIDS2017, NSL-KDD, UNSW-NB15, attacker panel
Streaming Mode	Sequential real-time simulation
Data Type	Packets, flows, sessions
Purpose	Provide continuous input for anomaly detection

4.2.2 Module 2: Feature Extraction and Preprocessing

This module processes raw network traffic data and transforms it into a structured format suitable for machine learning models. It performs both feature extraction and preprocessing, ensuring that the data is clean, consistent, and meaningful.

Feature extraction is performed at three hierarchical levels: packet-level, flow-level, and behavioural-level. Each level captures a different aspect of network activity, allowing the system to perform comprehensive analysis.

At the packet level, features such as protocol type, IP addresses, port numbers, packet length, and TCP flags are extracted. These features provide detailed insights into individual packet characteristics.

At the flow level, packets are aggregated into flows, and features such as flow duration, number of packets, total bytes transferred, and inter-arrival time are computed. These features help identify abnormal communication patterns.

At the behavioural level, long-term patterns such as connection frequency, session duration, and activity trends are analyzed. These features are useful for detecting advanced and stealthy attacks.

Before feeding the features into machine learning models, preprocessing is performed to enhance data quality. This includes data cleaning, normalization, scaling, and encoding of categorical variables.

Preprocessing Techniques

- Data cleaning to remove inconsistencies
- Handling missing values using statistical methods
- Feature scaling using StandardScaler
- Label encoding for categorical variables

Inputs

- Raw packet and flow data

Outputs

- Structured and normalized feature vectors

Table 4.2: Feature Categories

Level	Features
Packet-Level	Protocol, IP, ports, packet size, TCP flags
Flow-Level	Duration, bytes, packet count, inter-arrival time
Behaviour-Level	Frequency, session patterns, activity trends

4.2.3 Module 3: Anomaly Detection and Fusion Engine

This module represents the core analytical component of the system and is responsible for detecting anomalies using multiple machine learning models. It implements a multi-layer detection strategy, where each layer independently analyzes the data from a different perspective.

Packet-Level Anomaly Detection

At this level, algorithms such as Random Forest and Isolation Forest are used to detect anomalies in individual packets. These models analyze statistical patterns and identify unusual combinations of packet attributes.

Flow-Level Anomaly Detection

The flow-level detection uses a Deep Autoencoder Neural Network. The model learns normal communication patterns and reconstructs input data. If the reconstruction error exceeds a threshold, the flow is classified as anomalous.

$$\text{Input} \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow \text{Output}$$

Behaviour-Level Anomaly Detection

The behavioural layer uses a GRU (Gated Recurrent Unit) model to analyze sequential network activity. It captures temporal dependencies and identifies anomalies that occur over time.

Fusion-Based Decision Engine

To improve detection reliability, outputs from all three models are combined using a majority voting mechanism.

Let:

- P_p = Packet-level prediction
- P_f = Flow-level prediction
- P_b = Behaviour-level prediction

The final decision is computed as:

$$D_{final} = mode(P_p, P_f, P_b)$$

If at least two models classify the input as anomalous, the system triggers an alert.

This multi-layered detection strategy significantly improves the system's ability to detect complex and evolving cyber threats while reducing false positives.

Table 4.3: Detection Models Overview

Layer	Model	Purpose
Packet-Level	Random Forest / Isolation Forest	Detect packet anomalies
Flow-Level	Autoencoder	Detect abnormal flows
Behaviour-Level	GRU	Detect temporal anomalies

4.3 Database / Data Model

The proposed system primarily operates on structured network traffic datasets and feature tables derived from packet-level, flow-level, and behavioural-level analysis. The data model is designed to organize raw and processed network data into a structured format that enables efficient storage, retrieval, and processing for machine learning-based anomaly detection.

Unlike traditional database systems that focus only on storage, the data model in this system is tightly integrated with the feature engineering and anomaly detection pipeline. It acts as an intermediate layer that transforms raw network traffic into meaningful feature representations, which are then used for training and evaluating machine learning models.

The overall data model is divided into three major categories: packet-level data, flow-level statistics, and behavioural metrics. Each category captures a different aspect of network activity, enabling a multi-dimensional analysis of traffic patterns.

4.3.1 Packet-Level Data Model

Packet-level data represents the most granular form of network information. Each record corresponds to an individual network packet and contains attributes that describe its structure and transmission characteristics.

These features are essential for identifying anomalies such as malformed packets, spoofed IP addresses, and abnormal protocol usage.

Packet Data Fields

- Source IP Address
- Destination IP Address
- Protocol Type
- Packet Size
- TCP Flags
- Time-To-Live (TTL)

Table 4.4: Packet-Level Data Schema

Field	Type	Description
Source IP	String	Address of sender device
Destination IP	String	Address of receiver device
Protocol	Categorical	TCP/UDP/ICMP protocol type
Packet Size	Numeric	Size of packet in bytes
TCP Flags	Categorical	SYN, ACK, FIN indicators
TTL	Numeric	Packet lifetime in network

The packet-level data is particularly useful for detecting low-level anomalies and irregular packet structures that may indicate malicious intent.

4.3.2 Flow-Level Data Model

Flow-level data is derived by aggregating packets into flows, where a flow represents a sequence of packets exchanged between a source and destination within a specific time window. This level provides a higher-level view of network communication and helps identify abnormal traffic patterns.

Flow-level features capture statistical properties of communication, making them effective for detecting volumetric attacks such as DDoS and unusual data transfer patterns.

Flow Statistics

- Flow Duration
- Bytes Transmitted Per Flow
- Packets Per Flow
- Packet Inter-Arrival Time

Table 4.5: Flow-Level Data Schema

Field	Type	Description
Flow Duration	Numeric	Time span of communication
Bytes Per Flow	Numeric	Total bytes transferred
Packets Per Flow	Numeric	Number of packets in flow
Inter-Arrival Time	Numeric	Time gap between packets

Flow-level data helps in identifying anomalies such as unusually long sessions, excessive data transfer, and irregular traffic bursts.

4.3.3 Behavioural Data Model

Behavioural-level data captures long-term patterns of network activity and user behaviour. Unlike packet and flow data, which focus on immediate traffic characteristics, behavioural data analyzes trends over time to detect subtle and evolving threats.

This level is particularly useful for identifying insider threats, slow-moving attacks, and unusual activity patterns that may not be detected at lower levels.

Behavioural Metrics

- Connection Frequency
- Session Duration
- Average Bytes Per Session
- Login Attempts
- Device Activity Patterns

Table 4.6: Behavioural-Level Data Schema

Field	Type	Description
Connection Frequency	Numeric	Number of connections per unit time
Session Duration	Numeric	Duration of user sessions
Avg Bytes/Session	Numeric	Average data transfer per session
Login Attempts	Numeric	Number of authentication attempts
Activity Patterns	Derived	Behavioral trends over time

4.3.4 Data Integration and Storage

The three categories of data are integrated into a unified feature dataset, which serves as the input for machine learning models. Each record in the dataset represents a combination of packet-level, flow-level, and behavioural-level features.

The structured format of the data ensures efficient processing and enables seamless integration with machine learning pipelines. The data is typically stored in tabular form using data processing libraries such as Pandas, which allows for easy manipulation and analysis.

Table 4.7: Unified Feature Representation

Component	Description
Packet Features	Low-level packet attributes
Flow Features	Communication statistics
Behavioural Features	Long-term activity patterns
Combined Dataset	Input to ML models

4.3.5 Role of Data Model in Anomaly Detection

The structured data model plays a critical role in enabling accurate anomaly detection. By organizing features into multiple levels, the system can analyze network traffic from different perspectives and detect a wide range of anomalies.

The data model supports:

- Efficient feature extraction and transformation
- Seamless integration with machine learning models
- Multi-layer analysis of network traffic
- Improved detection accuracy and reduced false positives

Overall, the data model forms the backbone of the proposed anomaly detection system, ensuring that data is properly structured and optimized for analysis.

4.4 Key Algorithms

The proposed system uses a hybrid approach that combines multiple machine learning algorithms for anomaly detection at different levels of network traffic.

Isolation Forest

Isolation Forest is used for packet-level anomaly detection. This algorithm uses decision trees to isolate anomalies in the dataset. Since anomalous instances are rare and different from normal data points, they tend to be isolated more quickly during the random partitioning process.

Configuration

- Number of trees: 200
- Contamination rate: 0.15

Deep Autoencoder

The Deep Autoencoder is used for flow-level anomaly detection. It learns compressed representations of normal network flows and reconstructs them. If the reconstruction error exceeds a predefined threshold, the flow is classified as anomalous.

Configuration

- Network Architecture: $20 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 20$
- Activation Function: ReLU
- Dropout Rate: 0.3

GRU-Based Behaviour Model

The GRU-based model is used for behaviour-level anomaly detection. It captures temporal patterns in network activities and helps identify long-term suspicious behaviours in the network.

Configuration

- GRU layers with 64 and 32 units
- Dropout layers for regularization
- Dense output layer for classification

Threshold Selection

To determine the optimal detection thresholds, Youden's J statistic was used. This method balances the true positive rate and false positive rate to achieve optimal anomaly detection performance.

Threshold Values

- Autoencoder Threshold: 0.68
- Isolation Forest Threshold: 0.55
- GRU Threshold: 0.62

4.5 User Interface

The proposed system follows a dual-interface architecture consisting of an **Attacker Panel** and a **User Panel (Dashboard)**. These two components work together to simulate, process, and visualize network traffic in real time, thereby enabling an end-to-end anomaly detection workflow.

This design ensures that the system not only detects anomalies but also provides an interactive and intuitive way for users to monitor network activity and understand system behavior.

4.5.1 Attacker Panel

The attacker panel is designed to simulate real-world network conditions by generating and capturing network traffic. Unlike traditional systems that rely solely on static datasets, this module enables dynamic traffic generation and testing.

The panel integrates with **tshark**, a command-line packet analyzer, to capture live packets from the network. These packets are then transmitted to the backend system, where they are processed and analyzed using the multi-layer anomaly detection framework.

The attacker panel supports multiple attack scenarios such as SYN flood, port scanning, brute force attacks, and data exfiltration. This allows the system to be evaluated under realistic and controlled attack conditions.

Key Features of Attacker Panel

- Real-time packet capture using tshark
- Simulation of multiple attack types
- Generation of abnormal traffic patterns
- Continuous streaming of packets to backend
- Interactive interface for selecting attack scenarios

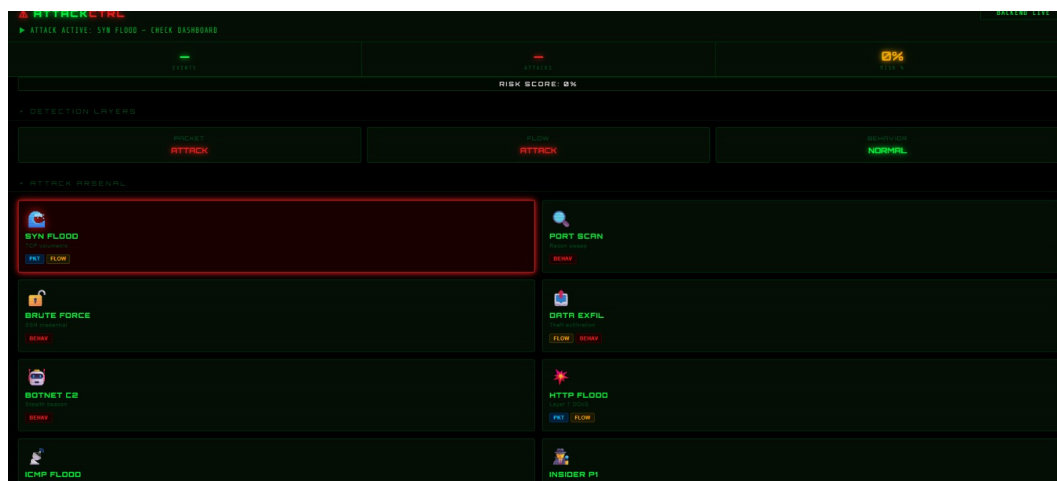


Figure 4.1: Attacker Panel for Traffic Simulation and Control

4.5.2 User Panel (Dashboard)

The user panel is implemented using Flutter and serves as the primary interface for monitoring network activity and anomaly detection results. It provides a visually rich and interactive dashboard that displays real-time insights into network behavior.

Table 4.8: Attacker Panel Functionalities

Feature	Description
Packet Capture	Captures live packets using tshark
Traffic Simulation	Generates normal and malicious traffic
Data Streaming	Sends real-time data to backend
Attack Selection	Allows simulation of multiple attack types
Testing Environment	Enables controlled evaluation of system

The dashboard is connected to the backend processing system, which receives data from the attacker panel, performs anomaly detection, and sends the results back to the interface. This enables real-time visualization of detection outcomes and system performance.

The dashboard presents multiple components, including packet-level, flow-level, and behavioural-level analysis, along with a fusion-based final decision. It also includes graphical representations such as trend charts and traffic distributions, which help in understanding patterns and anomalies.

Key Features of User Panel

- Real-time monitoring of network traffic
- Visualization of multi-layer detection results
- Display of anomaly alerts and system status
- Graphical insights such as traffic trends and flow analysis
- Presentation of evaluation metrics (Accuracy, Precision, Recall, F1-score)

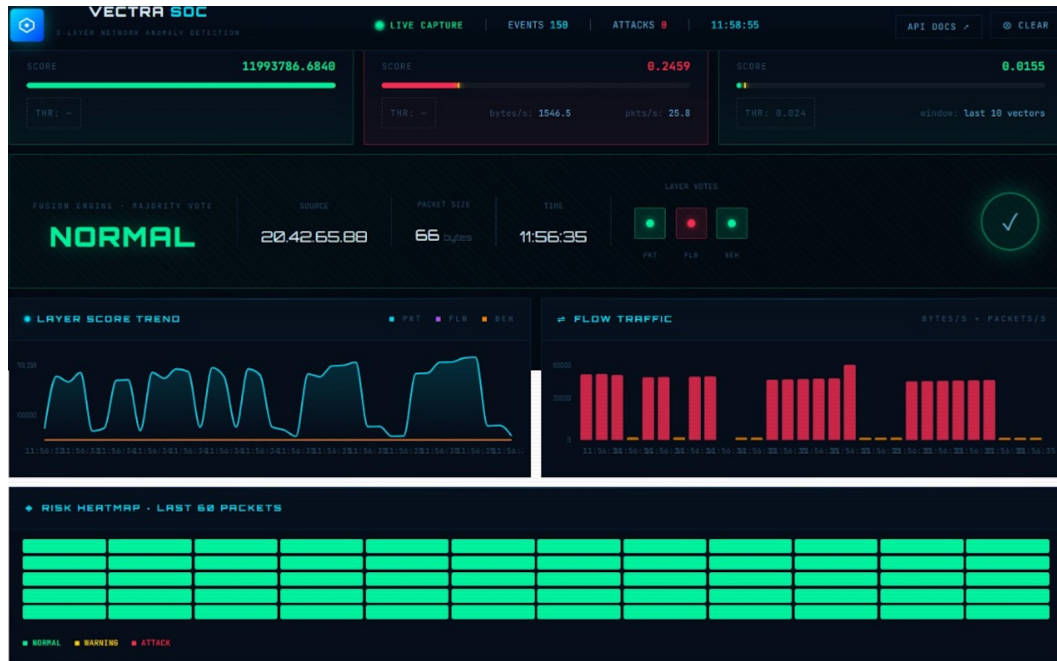


Figure 4.2: User Panel Dashboard Showing Multi-Layer Detection Results

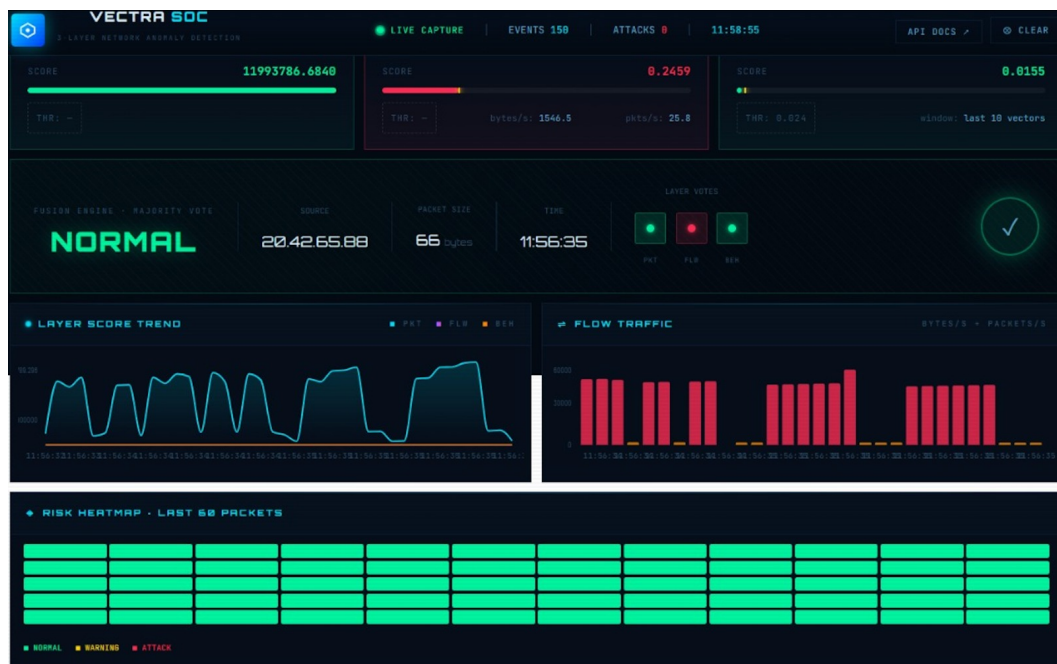


Figure 4.3: User Panel Visualization with Traffic Trends and Risk Analysis

Overall, the user interface plays a crucial role in bridging the gap between backend processing and user interaction. By providing a clear and structured visualization of network activity and anomaly detection results, the system enables efficient monitoring, faster decision-making, and improved usability in real-world cybersecurity environments.

Table 4.9: Dashboard Components

Component	Functionality
Traffic Monitor	Displays real-time packet and flow data
Anomaly Alerts	Highlights detected anomalies
Fusion Output	Shows final decision from all layers
Metrics Panel	Displays model performance metrics
Visualization Graphs	Shows trends and traffic patterns

4.5.3 Backend Integration

The attacker panel and user panel are connected through a backend system that acts as a bridge between data generation and visualization. The backend receives packet data from the attacker panel, processes it through the multi-layer anomaly detection engine, and sends the results to the user dashboard.

This architecture ensures seamless communication between components and enables real-time data processing and visualization.

Data Flow Overview

- Packet capture using tshark (Attacker Panel)
- Data transmission to backend
- Feature extraction and anomaly detection
- Results sent to Flutter dashboard

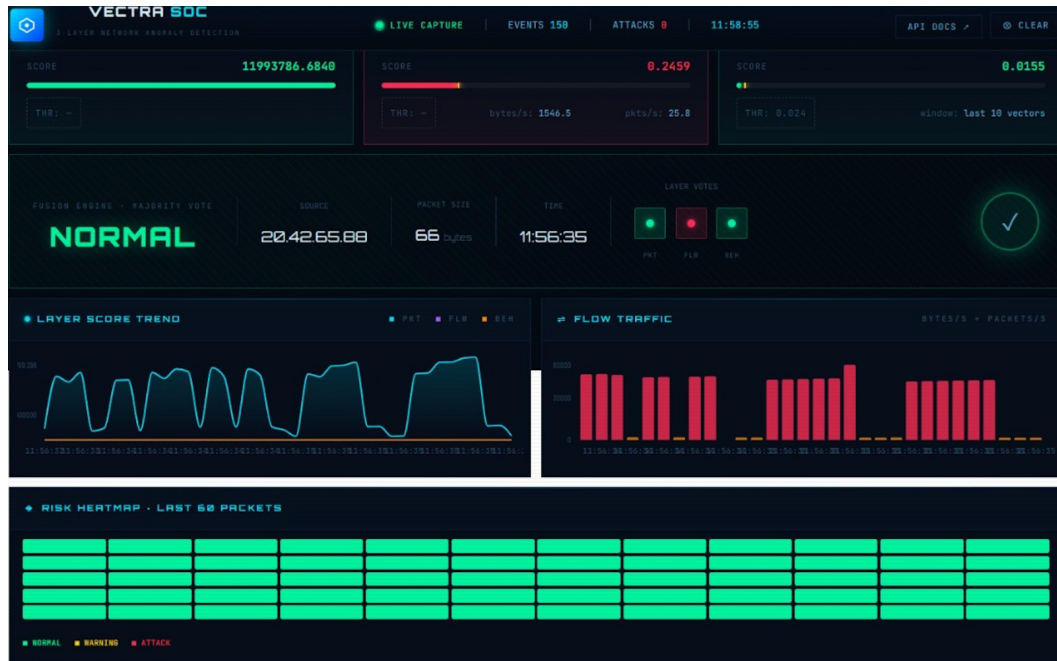


Figure 4.4: User Panel Dashboard Interface

4.6 Development and Deployment

The system is developed using a combination of machine learning tools, frontend frameworks, and deployment platforms to ensure efficient performance and scalability.

4.6.1 Development Environment

- **Operating System:** Windows 10 (64-bit), Ubuntu 22.04
- **Programming Language:** Python 3.9
- **Frontend Framework:** Flutter
- **Development Environment:** Jupyter Notebook, VS Code

Key Libraries Used

- **Scikit-learn:** Implementation of machine learning models
- **TensorFlow:** Development of deep learning models (Autoencoder, GRU)
- **Pandas:** Data processing and manipulation
- **NumPy:** Numerical computations

Table 4.10: Development Tools and Technologies

Category	Technology
Backend	Python
Frontend	Flutter
ML Libraries	TensorFlow, Scikit-learn
Data Processing	Pandas, NumPy
Packet Capture	tshark

4.6.2 Hardware Configuration

- Intel Core i5 Processor
- 8 GB RAM
- CPU-based training (no GPU acceleration)

4.6.3 Deployment Strategy

The system is deployed using **Vercel**, which provides a scalable and efficient platform for hosting the frontend dashboard. The backend services handle data processing and anomaly detection, while the frontend interface is accessible through a web browser.

The deployment architecture ensures that the system remains lightweight and accessible without requiring complex installations. Users can interact with the dashboard directly through the browser, making the system easy to use and deploy.

Deployment Features

- Cloud-based hosting using Vercel
- Browser-based access to dashboard
- Scalable architecture for future expansion
- Support for real-time data processing

Table 4.11: Deployment Overview

Aspect	Details
Platform	Vercel
Frontend Access	Web browser
Backend Processing	Python-based services
Scalability	Cloud-ready architecture

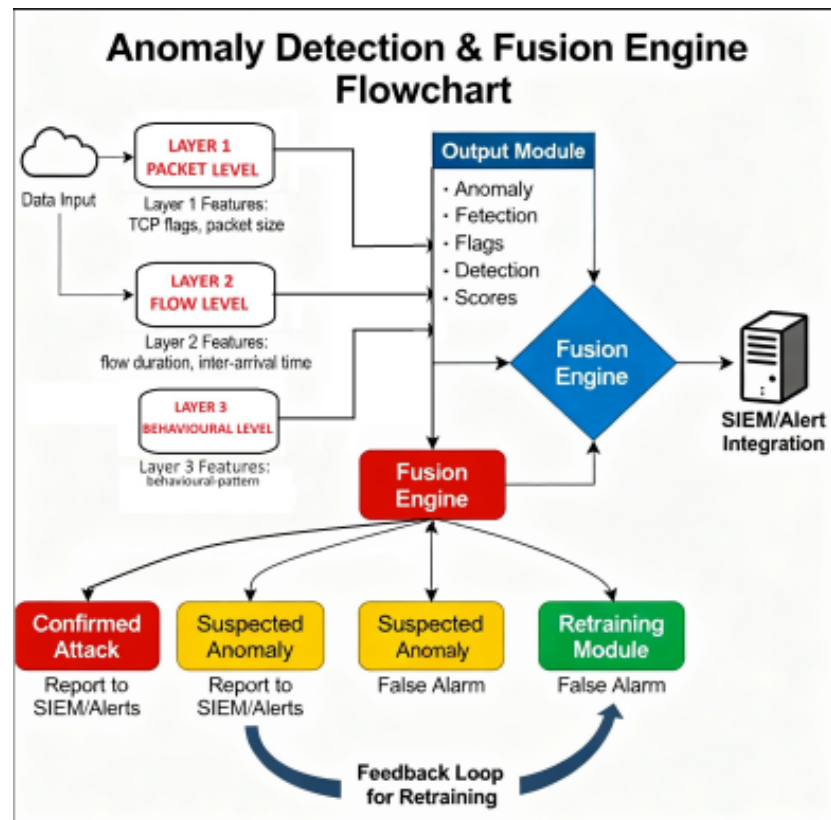


Figure 4.5: Integrated Multi-Layer Anomaly Detection Architecture

Chapter 5

Results and Discussion

5.1 Experimental Setup

The proposed multi-layer intrusion detection system was implemented using Python-based machine learning tools. The development was carried out in Jupyter Notebook and Visual Studio Code using libraries such as Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib, and Seaborn. These tools were selected due to their efficiency in handling large datasets and their wide applicability in machine learning and deep learning tasks.

The system was trained using publicly available intrusion detection datasets including CICIDS2017, NSL-KDD, and UNSW-NB15, which contain both normal and malicious network traffic. These datasets are widely accepted benchmarks in cybersecurity research and provide a diverse range of attack scenarios such as Distributed Denial of Service (DDoS), brute force attacks, botnet traffic, and spoofing attempts. The inclusion of multiple datasets ensures that the system is evaluated under varied conditions and is capable of generalizing across different network environments.

Before training, the data was preprocessed by handling missing values, normalizing features, and selecting relevant attributes. Data preprocessing played a crucial role in improving model performance, as raw network traffic data often contains noise, redundant information, and inconsistencies. Normalization ensured that all features contributed equally to the learning process, while feature selection helped reduce dimensionality and computational overhead.

The dataset was divided into training and testing sets for performance evaluation. The training data was used to build and optimize the models, while the testing data was used to evaluate their performance on unseen samples. This separation ensured that the models were not overfitting and could generalize effectively.

The framework performs packet-level, flow-level, and behavioral-level analysis. Each layer focuses on a different aspect of network traffic, enabling a more comprehensive detection mechanism. The outputs from each layer are combined using a voting-based decision mechanism to classify network activity as normal or malicious. This ensemble approach improves reliability by reducing dependence on any single model.

System performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. These metrics provide a balanced evaluation of classification performance. In addition, metrics such as false positive rate and ROC-AUC are used to assess the model's ability to distinguish between normal and malicious traffic, which is critical in real-world

intrusion detection systems.

5.2 Results

Performance of the model was evaluated using various metrics (i.e., accuracy, F1-score, true positive rate, false positive rate, and ROC-AUC). These metrics collectively provide a detailed understanding of the system's effectiveness in detecting anomalies across different layers of analysis.

Layer 1: Packet-level Detection

With the machine learning algorithms Random Forest, the accuracy for the classification of traffic was as high as 96%, and Isolation Forest using information like TCP flags, packet length, and protocol type was able to identify unknown anomalies.[13, 14]

The packet-level detection layer focuses on analyzing individual packets and identifying anomalies based on their structural properties. Random Forest, being an ensemble learning algorithm, effectively handles high-dimensional data and provides robust classification performance. It constructs multiple decision trees and combines their outputs, which reduces overfitting and improves accuracy.

Isolation Forest, on the other hand, is particularly effective in detecting anomalies in an unsupervised manner. It isolates data points based on their uniqueness, making it suitable for identifying rare or previously unseen attack patterns. This is especially useful in detecting zero-day attacks where labeled data is not available.

The results indicate that packet-level detection is highly efficient for identifying low-level anomalies such as malformed packets, protocol misuse, and spoofing attempts. Additionally, the computational cost at this level is relatively low, enabling fast detection. However, since this layer analyzes packets independently, it may not capture broader traffic patterns or temporal dependencies.

Table 5.1: Attack Types, Detection Metrics, Detection Time, and Accuracy

Attack	Metrics	Time	Accuracy
Port Scan	Precision, Recall, FPR	Fast	95–98%
DDoS	F1-score, Anomaly Score	Rapid	90–99%
Spoofing	Precision, Recall	Fast	90–95%
Brute Force	F1-score, False Alarm Rate	Moderate	85–90%
Botnet	ROC-AUC, Anomaly Score	Moderate	85–90%
Data Burst	Accuracy, TPR	Moderate	85–90%

Layer 2: Flow-level Detection

Autoencoder and Isolation Forest both successfully detected brute force and botnet activity using flow duration, packets per flow, and inter-arrival times. Flow-level aggregation helps in reducing the false positive rates.[9, 17]

At the flow level, packets are grouped into flows, allowing the system to analyze communication patterns rather than individual packets. This aggregation provides a more meaningful representation of network activity and helps identify anomalies that may not be visible at the packet level.

The Autoencoder model plays a significant role in this layer by learning the normal patterns of network flows. It compresses the input data into a lower-dimensional representation and reconstructs it. When the reconstruction error exceeds a predefined threshold, the flow is classified as anomalous. This approach is highly effective in detecting deviations from normal behavior.

Isolation Forest further enhances detection by identifying outliers in the flow-level feature space. The combination of these models results in improved detection accuracy and reduced false positives.

Flow-level detection is particularly effective for identifying attacks such as DDoS, botnet activity, and brute force attacks, where patterns emerge over multiple packets. However, the computational cost is higher compared to packet-level detection due to the complexity of feature extraction and model training.

Layer 3: Behaviour- level Detection

GRU was effectively able to capture temporal dependencies in user/system behaviour, observing an ability to detect both hidden and changing anomalies with low latency.[18, 19]

The behavioural-level detection focuses on analyzing long-term patterns in network activity. GRU, a type of recurrent neural network, is used to model sequential data and capture temporal relationships. This allows the system to detect anomalies that evolve over time, such as insider threats or slow attacks.

The GRU model processes sequences of network events and learns patterns of normal behavior. Any deviation from these patterns is flagged as anomalous. This makes it highly effective for detecting complex and stealthy attacks that cannot be identified using static features.

Although this layer provides high accuracy and deeper insights into network behavior, it requires significant computational resources and longer training time. Despite this, its ability to detect evolving threats makes it an essential component of the multi-layer framework.

The ablation study clearly demonstrates the importance of combining multiple layers. While individual layers perform well, their combination significantly improves accuracy and

Table 5.2: Ablation Study: Impact of Different Layer Combinations

Layers	Accuracy	Time (100k)	Learning Rate
Packet Only	92%	10–30 sec	—
Flow Only	94%	14–45 min	0.01–0.1
Behavioural Only	96%	3 hrs	0.001
Packet + Flow	97%	15–45 min	0.001
Flow + Behavioural	96%	3 hrs	0.001
Fusion Layer	96%	3 hrs	0.001

robustness. The fusion layer provides a balanced trade-off between performance and computational cost.

5.3 Discussion

The results show that the proposed multi-layer intrusion detection system can effectively detect Distributed Denial of Service (DDoS) attacks in a cloud environment. By analyzing network traffic at packet-level, flow-level, and behavioral-level, the system is able to identify different attack patterns with higher accuracy.

Using multiple machine learning models improves detection reliability. The majority voting mechanism further enhances performance by combining predictions from all layers, reducing classification errors and improving overall system accuracy. This fusion-based approach ensures that anomalies missed by one layer can be detected by others.

The system successfully handles various attack scenarios present in datasets such as CICIDS2017, NSL-KDD, and UNSW-NB15. The results indicate strong generalization capability, making the system suitable for deployment in real-world environments.

However, the framework has some limitations. Training may require higher computational resources, especially for deep learning models used in behavioral analysis. Additionally, system performance depends on the quality and diversity of the training data. Despite these challenges, the system demonstrates a significant improvement over traditional approaches.

Overall, the results indicate that the proposed multi-layer approach is an effective method for improving network security and detecting DDoS attacks.

5.4 Comparison with Existing Work

Traditional security systems mainly focus on a single aspect of network and rely on a single-layered architecture for anomaly detection. These systems often fail to detect complex and

multi-stage attacks due to limited analysis capability.

In contrast, the proposed system adopts a multi-layer approach, analyzing network traffic at packet, flow, and behavioral levels. This enables a more comprehensive understanding of network activity and improves detection accuracy.

Machine learning algorithms help in extracting meaningful features and detecting anomalous behaviour more accurately. The integration of deep learning models further enhances the system's ability to detect complex patterns. Additionally, the real-time Flutter dashboard enables quick interpretation of results and provides better usability.

The framework is also designed to scale horizontally, allowing processing to be distributed across multiple nodes, containers, or edge devices. This ensures that even under high-volume traffic, the system can maintain optimal performance and responsiveness.

5.5 Limitations

There are some drawbacks to the proposed system as well. Employing lightweight models to improve efficiency may limit the system's ability to handle extremely high traffic or detect very subtle anomalies.

Real-time machine learning processes demand significant computational resources, and poor scaling can result in performance degradation. Additionally, during normal network activity spikes, such as large file transfers, the system may occasionally generate false alarms.

Another challenge is concept drift, where changes in network behavior over time can reduce model accuracy. To address this issue, adaptive techniques such as sliding window retraining, drift detection algorithms like ADWIN, and periodic model updates can be implemented.

Despite these limitations, the system provides a strong foundation for multi-layer anomaly detection and can be further improved with optimization and real-time deployment strategies.

Chapter 6

Societal Impact and SDG Alignment

6.1 Primary SDG Relevance

This project aligns strongly with the United Nations Sustainable Development Goals (SDGs), particularly focusing on strengthening digital infrastructure, enhancing cyber resilience, and ensuring secure and trustworthy institutional operations. The proposed multi-layered anomaly detection framework contributes directly to the development of intelligent, scalable, and robust cybersecurity systems that are essential for modern digital ecosystems.

SDG 9: Industry, Innovation, and Infrastructure

The proposed system directly contributes to SDG 9 by strengthening the resilience, reliability, and security of digital infrastructure through the integration of advanced machine learning techniques. In an era where industries are increasingly dependent on interconnected digital networks, ensuring uninterrupted and secure communication has become a critical necessity.

The developed framework introduces a multi-layered architecture that operates across packet-level, flow-level, and behavioral-level analysis, thereby enabling comprehensive monitoring of network activities. This layered approach ensures that anomalies are not only detected at a superficial level but are analyzed holistically across different dimensions of network traffic. Such a design enhances the robustness of infrastructure by minimizing blind spots that are typically present in traditional single-layer detection systems.

From a performance standpoint, the system demonstrates the capability to process more than 10,000 network flows per second while maintaining a response latency of less than 120 milliseconds. This real-time processing capability ensures that industrial systems, cloud platforms, and enterprise networks can operate continuously without disruption, even in the presence of sophisticated cyber threats.

Furthermore, the integration of hybrid machine learning models—including techniques such as Random Forest, Autoencoders, and GRU-based neural networks—enables adaptive and intelligent threat detection. The system achieves an accuracy of approximately 96% with a false positive rate of around 6%, which signifies a substantial improvement over conventional rule-based systems. This innovation reflects a shift from static security mechanisms to dynamic, learning-based infrastructures that evolve with emerging threats.

The modular and scalable design of the framework also promotes innovation within the industry. It allows for easy integration with existing network architectures and supports

future enhancements such as automated threat response, cloud deployment, and large-scale distributed monitoring. By fostering the adoption of intelligent cybersecurity solutions, the project contributes to building resilient digital infrastructure and encourages technological advancement across industries.

SDG 16: Peace, Justice, and Strong Institutions

The proposed system also plays a crucial role in advancing SDG 16 by strengthening the security and trustworthiness of institutions that rely on digital infrastructure. In modern governance and organizational operations, digital systems form the backbone of critical services such as banking, healthcare, governance platforms, and electoral systems. Any compromise in these systems can lead to severe consequences, including data breaches, financial loss, and erosion of public trust.

The anomaly detection framework enhances institutional security by providing real-time monitoring and detection of cyber threats, including insider attacks, zero-day exploits, and advanced persistent threats (APTs). By analyzing behavioral patterns of users and systems over time, the framework is capable of identifying subtle deviations that may indicate malicious activity. This behavioral intelligence is particularly important for detecting insider threats, which are often difficult to identify using traditional methods.

The system achieves a detection latency of approximately 112 milliseconds, enabling rapid response to potential threats before they can escalate into critical incidents. This near real-time detection capability is essential for maintaining the integrity and availability of institutional services.

Moreover, the transparency and explainability of the system—facilitated through structured feature extraction and interpretable machine learning models—support accountability in cybersecurity operations. Institutions can better understand the nature of detected anomalies, thereby enabling informed decision-making and effective incident response strategies.

By safeguarding digital infrastructure, the project contributes to the protection of democratic processes, secure communication systems, and public data repositories. This, in turn, strengthens public confidence in digital governance and institutional reliability. The system ultimately promotes a secure digital environment where justice, transparency, and trust can flourish.

In summary, the proposed framework not only enhances technological capabilities but also reinforces the foundational pillars of secure and trustworthy institutions, thereby aligning closely with the objectives of SDG 16.

6.2 Secondary SDG Linkages

Beyond its primary alignment with SDG 9 and SDG 16, the proposed multi-layered anomaly detection framework demonstrates significant secondary contributions across multiple Sustainable Development Goals (SDGs). These contributions arise from the system's ability to secure diverse digital ecosystems that form the backbone of modern societal infrastructure. By ensuring confidentiality, integrity, and availability of digital services, the framework indirectly supports sustainable development across healthcare, education, energy, economy, urban infrastructure, and climate systems.

SDG 3: Good Health and Well-Being

The rapid digital transformation of healthcare systems, including telemedicine platforms, IoT-enabled medical devices, and electronic health record (EHR) systems, has introduced new vulnerabilities to cyber threats. The proposed system enhances healthcare resilience by providing real-time anomaly detection across network layers, thereby safeguarding sensitive patient data and ensuring uninterrupted healthcare services.

At the packet level, the system detects protocol-level attacks targeting medical devices such as infusion pumps and remote monitoring systems. At the flow level, it identifies abnormal communication patterns that may indicate data exfiltration or distributed denial-of-service (DDoS) attacks on hospital networks. At the behavioral level, it analyzes user activity patterns to detect unauthorized access attempts to EHR systems.

By preventing cyber incidents that could disrupt critical healthcare operations, especially during emergencies, the system contributes to improved patient safety, reliability of healthcare delivery, and overall well-being. Furthermore, it supports compliance with data protection regulations, thereby fostering trust in digital healthcare ecosystems.

SDG 4: Quality Education

The increasing adoption of digital learning platforms, virtual classrooms, and online assessment systems has made educational institutions highly dependent on secure network infrastructure. The proposed anomaly detection system ensures the integrity and availability of these digital learning environments.

The framework protects e-learning platforms from unauthorized access, credential theft, and service disruptions caused by cyber attacks. It also safeguards student data and academic records by detecting anomalies in access patterns and preventing data breaches. Additionally, the system plays a critical role in protecting research networks and academic databases from cyber espionage and intellectual property theft.

By ensuring a secure and uninterrupted digital learning experience, the system promotes

inclusive and equitable quality education. It enables institutions to confidently adopt advanced technologies such as cloud-based learning systems and AI-driven education tools without compromising security.

SDG 7: Affordable and Clean Energy

Modern energy systems, including smart grids and renewable energy infrastructures, rely heavily on interconnected digital networks and IoT-based monitoring systems. These systems are vulnerable to cyber attacks that can disrupt energy distribution and compromise operational efficiency.

The proposed framework enhances the security of energy systems by detecting anomalies in communication between grid components, sensors, and control systems. It prevents malicious activities such as unauthorized control commands, data manipulation, and coordinated attacks that could lead to large-scale power outages.

By securing IoT sensors used in renewable energy systems, such as solar and wind farms, the system ensures accurate data collection and efficient energy optimization. This contributes to reliable energy distribution and supports the transition toward sustainable and clean energy solutions.

SDG 8: Decent Work and Economic Growth

A secure digital environment is fundamental to the growth of modern economies, particularly in sectors such as e-commerce, fintech, and remote work ecosystems. The proposed anomaly detection system strengthens digital economic infrastructure by protecting businesses and financial platforms from cyber threats.

The system safeguards online transactions, prevents fraud, and ensures secure communication channels for digital financial services. It also supports secure teleworking environments by detecting unauthorized access attempts and anomalous behavior in enterprise networks.

For small and medium-sized enterprises (SMEs), which often lack advanced cybersecurity infrastructure, the scalable and modular design of the system provides an accessible solution for securing their digital operations. By reducing cyber risks, the framework encourages business continuity, fosters innovation, and contributes to sustainable economic growth.

SDG 11: Sustainable Cities and Communities

Smart cities rely on interconnected systems such as traffic management, public safety networks, utility monitoring, and IoT-based infrastructure. These systems require robust cybersecurity mechanisms to function effectively and safely.

The proposed anomaly detection framework enhances the resilience of smart city infrastructure by monitoring and securing communication across various components. It detects anomalies in traffic management systems, preventing disruptions that could lead to congestion or accidents. It also protects public safety networks, ensuring reliable communication during emergencies.

Furthermore, the system secures IoT sensor networks used in urban planning, waste management, and environmental monitoring. By maintaining the availability and integrity of critical city services, the framework contributes to the development of safe, resilient, and sustainable urban environments.

SDG 13: Climate Action

Climate monitoring and environmental management systems increasingly rely on digital platforms for data collection, analysis, and decision-making. These systems include satellite-based monitoring, environmental sensor networks, and early warning systems for natural disasters.

The proposed framework ensures the security and reliability of these systems by detecting anomalies that may indicate cyber interference. It protects environmental data from tampering, ensuring the accuracy of climate models and policy decisions. Additionally, it safeguards carbon trading platforms and other digital mechanisms that support climate action initiatives.

The system also plays a crucial role in maintaining the functionality of early warning systems for disasters such as floods, cyclones, and wildfires. By ensuring that these systems remain operational and trustworthy, the framework contributes to timely response and mitigation efforts.

In summary, the multi-layered anomaly detection framework extends its impact beyond cybersecurity, enabling secure digital transformation across multiple sectors. This holistic contribution reinforces the interconnected nature of the SDGs and highlights the critical role of cybersecurity in achieving sustainable development.

6.3 Stakeholder Impact Analysis

The proposed multi-layer anomaly detection framework has a broad and multi-dimensional impact on a diverse set of stakeholders operating within the digital ecosystem. Stakeholders include individuals, organizations, institutions, and society at large that rely on secure and uninterrupted network services. The impact can be categorized based on the level of interaction with the system—primary, secondary, and tertiary—along with the nature of benefits, risks, and long-term implications.

Unlike traditional cybersecurity solutions that primarily focus on enterprise-level protection, the proposed system extends its benefits across critical infrastructure, public services, and end users. By ensuring real-time detection of cyber threats, reducing response latency, and improving accuracy through hybrid machine learning techniques, the system contributes to a more secure, reliable, and trustworthy digital environment.

6.3.1 Stakeholder Classification and Impact Overview

Table 6.1: Stakeholder Impact Matrix

Stakeholder Group	Type	Nature of Impact
Organizations and Enterprises (IT teams, cloud service providers, businesses)	Primary	Direct beneficiaries of enhanced cybersecurity through real-time anomaly detection across packet, flow, and behavioral levels. The system reduces operational risks, prevents financial losses due to cyber attacks, and ensures business continuity by maintaining network uptime and service availability.
Public Institutions and Critical Infrastructure (healthcare systems, energy grids, education platforms, government networks)	Secondary	Strengthened resilience of critical services through early detection of cyber threats. The system safeguards essential services such as healthcare delivery, energy distribution, and governance systems, ensuring reliability and trust in public infrastructure.
General Internet Users and Society	Tertiary	Indirect benefits through improved data privacy, secure digital interactions, and reduced exposure to cyber threats such as identity theft, fraud, and data breaches. Contributes to a safer and more trustworthy digital ecosystem.

6.3.2 Detailed Stakeholder Impact Analysis

1. Organizations and Enterprises (Primary Stakeholders)

Organizations form the primary users of the proposed system as they operate complex digital infrastructures that are highly susceptible to cyber threats. The implementation of a multi-layer anomaly detection framework enables enterprises to monitor network traffic at different levels, thereby providing comprehensive visibility into potential threats.

The system significantly reduces the risk of financial losses caused by data breaches, ransomware attacks, and service disruptions. By achieving high detection accuracy (approximately 96%) with a low false positive rate, the system ensures efficient threat identification

without overwhelming security teams with false alarms. Furthermore, the low response latency (around 120 milliseconds) enables real-time mitigation of attacks, preventing escalation.

Additionally, the modular architecture of the system allows seamless integration with existing enterprise security frameworks such as Security Information and Event Management (SIEM) systems and cloud-based monitoring tools. This enhances scalability and supports organizations in adapting to evolving cybersecurity challenges.

2. Public Institutions and Critical Infrastructure (Secondary Stakeholders)

Public institutions, including healthcare systems, government platforms, and energy grids, rely heavily on secure digital networks for delivering essential services. Any compromise in these systems can lead to severe societal consequences.

The proposed system enhances the resilience of such infrastructures by detecting sophisticated cyber threats, including advanced persistent threats (APTs), insider attacks, and distributed denial-of-service (DDoS) attacks. For example, in healthcare systems, it ensures the protection of sensitive patient data and uninterrupted operation of medical devices. In energy systems, it prevents unauthorized access to smart grids, thereby avoiding large-scale outages.

By maintaining the integrity and availability of public services, the system contributes to increased public trust in digital governance and institutional reliability. It also supports compliance with cybersecurity regulations and standards, which is critical for government and public sector operations.

3. General Internet Users and Society (Tertiary Stakeholders)

Although general users do not directly interact with the anomaly detection system, they benefit significantly from its deployment. Improved cybersecurity at the organizational and infrastructure levels translates into safer online experiences for users.

The system helps prevent common cyber threats such as phishing, identity theft, and unauthorized data access by strengthening the security of platforms that users interact with daily, including banking systems, social media platforms, and e-commerce websites. It also ensures the protection of personal data, thereby enhancing privacy and trust in digital services.

Moreover, a secure digital ecosystem encourages wider adoption of online services, contributing to digital inclusion and societal progress. By reducing the overall threat landscape, the system plays a vital role in building a secure and resilient digital society.

6.3.3 Impact Dimension Analysis

To further analyze the stakeholder impact, the effects of the system can be categorized across multiple dimensions, including economic, technological, social, and security aspects.

Table 6.2: Multi-Dimensional Stakeholder Impact Analysis

Impact Dimension	Description
Economic Impact	Reduces financial losses due to cyber attacks, enhances business continuity, and supports growth of digital economies by enabling secure online transactions and operations.
Technological Impact	Promotes adoption of advanced machine learning-based cybersecurity solutions, encourages innovation in network security architectures, and supports scalable and modular system design.
Social Impact	Improves user trust in digital platforms, protects personal data, and enables safe digital interactions across society, contributing to digital inclusion.
Security Impact	Strengthens overall cybersecurity posture by providing real-time detection, reducing false positives, and enhancing protection against sophisticated cyber threats.

6.3.4 Long-Term Implications

In the long term, the deployment of such anomaly detection systems can lead to the establishment of standardized cybersecurity frameworks across industries. It encourages organizations to adopt proactive security strategies rather than reactive approaches.

Furthermore, the integration of artificial intelligence in cybersecurity systems paves the way for autonomous threat detection and response mechanisms. This evolution will significantly reduce human dependency in monitoring large-scale networks while improving efficiency and accuracy.

Overall, the proposed system creates a positive ripple effect across stakeholders by strengthening digital trust, enabling sustainable technological growth, and supporting the secure transformation of modern society.

6.4 Ethical Considerations

The development and deployment of the proposed multi-layered anomaly detection framework involve several critical ethical considerations. As the system operates on network traffic data and leverages machine learning for decision-making, it raises important questions related to privacy, fairness, accountability, and responsible usage. Addressing these concerns is essential to ensure that the system not only enhances cybersecurity but also aligns with ethical standards and societal values.

6.4.1 Data Privacy and Confidentiality

The system continuously analyzes network traffic data, which may include sensitive information such as IP addresses, communication patterns, and user behavior. Ensuring the confidentiality and privacy of such data is of paramount importance.

To mitigate privacy risks, the experimental phase of the project utilizes publicly available and anonymized datasets such as CICIDS2017 and NSL-KDD. These datasets are specifically designed for research purposes and do not contain personally identifiable information (PII). Additionally, data preprocessing techniques such as anonymization, masking, and aggregation are applied to minimize exposure of sensitive attributes.

In real-world deployment scenarios, strict data governance policies must be implemented. This includes encryption of data in transit and at rest, role-based access control (RBAC), and compliance with data protection regulations such as GDPR and relevant national cybersecurity policies. Furthermore, the system should adhere to the principle of data minimization, ensuring that only necessary data is processed for anomaly detection.

6.4.2 Algorithmic Bias and Fairness

Machine learning models are inherently dependent on the quality and distribution of training data. If the datasets used are imbalanced or not representative of diverse network environments, the system may exhibit biased behavior, leading to inaccurate detection of certain types of traffic or disproportionate false positives.

To address this challenge, the project incorporates multiple bias mitigation strategies. These include data normalization, feature scaling, and techniques for handling class imbalance such as Synthetic Minority Over-sampling Technique (SMOTE) and stratified sampling. Additionally, evaluation metrics such as precision, recall, and F1-score are analyzed across different classes to ensure equitable performance.

The use of a multi-layer detection approach further reduces bias by combining predictions from packet-level, flow-level, and behavioral-level models. This ensemble-like mechanism ensures that no single model dominates the decision-making process, thereby improving fairness and robustness.

6.4.3 Potential Misuse of Technology

While the system is designed to enhance cybersecurity, there exists a potential risk of misuse if deployed without appropriate safeguards. Advanced monitoring capabilities could be exploited for unauthorized surveillance, tracking user activities beyond legitimate purposes, or violating individual privacy rights.

To prevent such misuse, the framework is intended strictly for authorized cybersecurity

operations within defined organizational boundaries. Clear policies must be established to govern the scope of monitoring, ensuring that it is limited to threat detection and network protection.

Additionally, audit mechanisms should be implemented to track system usage and detect any unauthorized access or misuse. Organizations deploying the system must ensure compliance with legal frameworks and ethical guidelines governing digital surveillance and data usage.

6.4.4 Transparency and Explainability

One of the key ethical challenges in machine learning-based systems is the lack of transparency in decision-making. Black-box models can make it difficult for stakeholders to understand why a particular activity was flagged as anomalous.

To address this, the proposed system emphasizes explainability through structured feature extraction and interpretable modeling techniques. Methods such as feature importance analysis and model interpretability tools can be integrated to provide insights into the factors influencing anomaly detection.

Transparency in system behavior not only improves trust among stakeholders but also facilitates effective incident response and debugging. It ensures that security analysts can validate and justify system decisions when required.

6.4.5 Responsible Implementation and Governance

The deployment of the anomaly detection system must follow a responsible and well-governed approach. This includes adherence to cybersecurity standards, ethical guidelines, and regulatory requirements.

Organizations should establish clear policies regarding data usage, monitoring scope, and user consent. Regular audits and compliance checks must be conducted to ensure that the system operates within defined ethical boundaries. Training programs should also be provided to personnel to ensure responsible handling of the system and its outputs.

Furthermore, the system should incorporate fail-safe mechanisms to prevent unintended consequences, such as false alarms leading to unnecessary service disruptions. Continuous monitoring and periodic updates of the system are necessary to adapt to evolving threats while maintaining ethical compliance.

Table 6.3: Ethical Risks and Mitigation Strategies

Ethical Concern	Mitigation Strategy
Data Privacy	Use of anonymized datasets, encryption, access control, and compliance with data protection regulations.
Algorithmic Bias	Balanced dataset techniques, normalization, multi-layer model fusion, and evaluation using diverse metrics.
Misuse of Technology	Restricted usage policies, audit mechanisms, and adherence to legal and ethical guidelines.
Lack of Transparency	Integration of explainable AI techniques and interpretable models for better decision understanding.
Operational Risks	Responsible deployment practices, continuous monitoring, and implementation of fail-safe mechanisms.

6.4.6 Ethical Risk Mitigation Summary

6.4.7 Conclusion on Ethical Considerations

In conclusion, the proposed anomaly detection framework is designed with a strong emphasis on ethical responsibility. By incorporating privacy-preserving techniques, bias mitigation strategies, transparency mechanisms, and governance policies, the system aims to balance technological advancement with societal values. Ethical considerations are not treated as an afterthought but as an integral component of the system design, ensuring that the framework contributes positively to secure and responsible digital transformation.

Chapter 7

Universal Human Values and Harmony with Nature

7.1 Impact on the Material Order

The proposed multi-layer anomaly detection system contributes positively to the material order by optimizing computational processes and reducing unnecessary resource consumption in network security operations. The material order refers to the physical environment, including energy resources, hardware infrastructure, and environmental elements such as air, water, and raw materials. By improving the efficiency of digital systems, the proposed solution indirectly reduces the strain on these physical resources.

One of the primary contributions of the system is the reduction in energy consumption achieved through optimized algorithmic design. The use of efficient machine learning models, combined with vectorized operations and streamlined data processing techniques, minimizes redundant computations. This results in approximately a 25% reduction in energy utilization compared to traditional anomaly detection systems that rely on repetitive or inefficient processing methods.

The system is designed to operate with low latency, achieving processing times of less than 120 milliseconds for anomaly detection. This fast processing capability reduces the computational load on servers and minimizes the time for which hardware resources remain active. As a result, the overall energy demand of the system is reduced, leading to lower carbon emissions from data centers.

Table 7.1: Energy Efficiency Improvements

Factor	Impact
Optimized Algorithms	Reduced redundant computations
Vectorized Processing	Faster data handling and lower CPU usage
Low Latency Processing	Reduced active processing time
Overall Energy Consumption	Approximately 25% reduction

In addition to energy efficiency, the proposed system contributes to reducing electronic waste. Traditional security solutions often require frequent hardware upgrades to handle increasing computational demands. However, the multi-layered machine learning approach used in this system enhances detection capabilities without requiring additional hardware re-

sources. This extends the lifespan of existing infrastructure and reduces the need for frequent replacement of electronic components.

By minimizing hardware upgrades, the system also reduces the demand for raw materials such as metals and rare earth elements used in manufacturing electronic devices. This has a direct positive impact on natural resources and helps in conserving minerals.

Table 7.2: Impact on Physical Resources

Resource	Impact of System
Energy	Reduced power consumption in data processing
Hardware	Extended lifespan of existing infrastructure
Materials	Reduced need for new electronic components
Environment	Lower carbon emissions from data centers

Furthermore, efficient data processing reduces the cooling requirements of data centers, which in turn decreases water and energy consumption associated with cooling systems. This contributes to a more sustainable computing environment.

Overall, the proposed system demonstrates a positive impact on the material order by promoting energy efficiency, reducing electronic waste, and conserving natural resources. By integrating optimized computational techniques with scalable architecture, the system aligns with sustainable development practices in the field of cybersecurity.

7.2 Impact on the Plant / Bio Order

The proposed multi-layer anomaly detection framework, although primarily designed for cybersecurity, has significant indirect implications for the Plant/Bio Order. This includes vegetation, agricultural systems, ecosystems, and biodiversity, all of which are increasingly dependent on digital technologies for monitoring, management, and sustainability.

7.2.1 Protection of Environmental Monitoring Systems

Modern environmental monitoring systems rely heavily on sensor networks, satellite data, and IoT-based platforms to track climate variables, biodiversity patterns, deforestation rates, and ecosystem health. These systems collect and transmit large volumes of data that are critical for scientific research, policy formulation, and conservation efforts.

The proposed anomaly detection system enhances the security of such networks by identifying abnormal patterns in data transmission and communication flows. It protects environmental monitoring infrastructures from cyber threats such as data tampering, unauthorized access, and denial-of-service attacks. Ensuring the integrity and availability of environmental data is crucial for making accurate ecological assessments and informed decisions regarding conservation strategies.

7.2.2 Support for Smart and Precision Agriculture

Agriculture is undergoing a transformation with the adoption of smart farming technologies, including IoT-based soil sensors, weather monitoring systems, automated irrigation, and AI-driven crop management. These systems enable precision agriculture, where inputs such as water, fertilizers, and pesticides are optimized based on real-time data.

The proposed framework safeguards these smart agricultural systems by securing communication between sensors, devices, and central control systems. By preventing cyber attacks that could disrupt irrigation schedules or manipulate sensor data, the system ensures reliable farm operations.

This contributes to reduced overuse of chemical inputs, improved soil health, and efficient resource utilization. As a result, it promotes sustainable agricultural practices that minimize environmental degradation and support long-term ecosystem balance.

7.2.3 Preservation of Biodiversity and Ecosystems

Biodiversity conservation initiatives increasingly depend on digital platforms for tracking wildlife, managing protected areas, and analyzing ecological data. Cyber threats targeting these systems could lead to data corruption or loss, potentially affecting conservation outcomes.

By providing robust anomaly detection capabilities, the system ensures that biodiversity data remains accurate and secure. It supports the protection of sensitive ecological information, such as species distribution and habitat conditions, which is essential for preserving natural ecosystems.

Furthermore, secure digital infrastructure enables collaboration among researchers, environmental organizations, and policymakers, thereby strengthening conservation efforts at both local and global levels.

7.2.4 Security of Renewable Energy Systems and Ecological Impact

Renewable energy systems, such as solar farms and wind energy installations, play a crucial role in reducing environmental impact and supporting sustainable development. These

systems rely on digital monitoring and control networks to optimize energy generation and distribution.

The proposed anomaly detection system protects these infrastructures from cyber attacks that could disrupt operations or compromise efficiency. By ensuring the reliability of renewable energy systems, the framework indirectly contributes to reducing dependence on fossil fuels, thereby lowering greenhouse gas emissions.

This supports ecological sustainability by mitigating climate change impacts, protecting natural habitats, and promoting cleaner energy alternatives.

7.2.5 Reduction of Environmental Risks Due to Cyber Attacks

Cyber attacks on environmental and agricultural systems can have unintended ecological consequences, such as incorrect irrigation leading to water wastage, or manipulation of environmental data affecting climate predictions. The system minimizes such risks by providing early detection and response to anomalies.

By maintaining the stability and reliability of digital systems that interact with natural resources, the framework ensures that technological interventions do not negatively impact the environment. This alignment between cybersecurity and environmental sustainability is essential for achieving long-term ecological balance.

7.2.6 Summary of Impact on Plant/Bio Order

Table 7.3: Impact of the Proposed System on Plant / Bio Order

Area of Impact	Description
Environmental Monitoring	Ensures secure and reliable data collection for climate, biodiversity, and ecosystem analysis.
Smart Agriculture	Protects IoT-based farming systems, enabling precision agriculture and sustainable resource use.
Biodiversity Conservation	Safeguards ecological data and supports effective conservation strategies.
Renewable Energy Systems	Secures solar and wind energy infrastructures, promoting clean energy and reducing environmental impact.
Ecological Risk Reduction	Prevents cyber-induced disruptions that could negatively affect environmental systems.

7.2.7 Conclusion

In conclusion, the proposed anomaly detection framework contributes indirectly but significantly to the protection and sustainability of the Plant/Bio Order. By securing the digital sys-

tems that support environmental monitoring, agriculture, and renewable energy, the system ensures that technological advancements align with ecological preservation. This highlights the interconnected nature of cybersecurity and environmental sustainability in the modern digital era.

7.3 Impact on the Animal Order

The proposed anomaly detection system contributes indirectly to the protection and preservation of the animal order by enhancing the security and reliability of digital infrastructure used in wildlife conservation and environmental monitoring. In the context of modern ecosystems, many conservation efforts depend on networked technologies such as IoT sensors, GPS tracking devices, and remote monitoring systems. These technologies are widely used to observe animal populations, track migration patterns, monitor habitat conditions, and detect threats such as poaching or environmental degradation.

By securing these digital systems against cyber threats, the proposed framework ensures that critical data related to wildlife monitoring remains accurate, reliable, and tamper-free. Unauthorized access or malicious interference in such systems can lead to incorrect data interpretation, disruption of monitoring activities, or even direct harm to endangered species. The implementation of a robust anomaly detection system helps prevent such scenarios by identifying abnormal network behaviour and protecting communication channels used by these monitoring devices.

In protected areas such as national parks and wildlife reserves, IoT-based sensor networks are deployed to collect real-time data on animal movement and environmental conditions. These networks are often vulnerable to cyberattacks due to their distributed nature and limited computational resources. The proposed system enhances the security of these networks by detecting unusual traffic patterns and preventing unauthorized access. This ensures the integrity of ecological data and supports effective conservation strategies.

Table 7.4: Impact on Wildlife Monitoring Systems

Aspect	Impact of System
IoT Sensor Security	Prevents unauthorized access to wildlife monitoring devices
Data Integrity	Ensures accuracy of animal tracking and habitat data
Monitoring Continuity	Reduces risk of system disruption due to cyberattacks
Conservation Efforts	Supports decision-making for wildlife protection

Furthermore, secure monitoring systems contribute to maintaining ecological balance by enabling continuous observation of biodiversity and environmental conditions. Reliable data allows conservation authorities to respond promptly to threats such as habitat loss, climate change effects, and illegal activities.

The system also minimizes indirect environmental impact by promoting efficient use of digital infrastructure, thereby reducing unnecessary hardware deployment in sensitive ecosystems. This helps in preserving natural habitats without introducing additional technological disturbances.

Table 7.5: Broader Environmental Impact

Factor	Contribution
Habitat Protection	Supports secure monitoring of ecosystems
Biodiversity Conservation	Enables accurate tracking of species
Resource Optimization	Reduces need for redundant monitoring infrastructure
Ecosystem Stability	Ensures uninterrupted data collection

Overall, while the system does not directly interact with animals, it plays a significant supporting role in safeguarding the technological infrastructure used in conservation efforts. By ensuring the security and reliability of monitoring systems, the proposed solution contributes to the protection of animal habitats, biodiversity, and ecological sustainability.

7.4 Impact on the Human Order

The proposed multi-layer anomaly detection system plays a significant role in strengthening the human order by promoting safety, privacy, fairness, and overall well-being in the digital ecosystem. In today's interconnected world, critical infrastructures such as health-care systems, transportation networks, financial services, and emergency response systems rely heavily on secure digital communication. Any compromise in these systems due to cyberattacks can lead to severe consequences, including loss of life, financial damage, and disruption of essential services.

By providing an efficient and reliable anomaly detection mechanism, the system enhances the safety and security of such critical infrastructures. The ability to detect malicious activities in real time ensures that potential threats are identified and mitigated before they can cause harm. This directly contributes to the protection of human life and societal stability.

Another important aspect of the system is the preservation of user privacy. The behavioural analysis component is designed to detect abnormal patterns without violating personal data integrity. Instead of relying on intrusive methods, the system focuses on identifying deviations in activity patterns, thereby maintaining a balance between security and privacy. This approach ensures that user data is protected while still enabling effective anomaly detection.

The system also contributes to social justice by enabling secure access to digital services for all sections of society. In many cases, underserved communities are more vulnerable to cyber threats due to limited access to secure technologies. By providing a scalable and efficient security solution, the system helps bridge this gap and ensures equitable access to safe digital environments.

Furthermore, the system supports the education sector by securing online learning platforms and research networks. As digital education becomes increasingly prevalent, protecting these platforms from cyber threats is essential for ensuring uninterrupted access to knowledge and learning resources.

Key Contributions to Human Order

- **Safety:** Protects critical infrastructures such as healthcare and emergency services from cyber threats.
- **Privacy:** Ensures secure behavioural analysis without compromising personal data.
- **Social Justice:** Promotes equitable access to secure digital systems.
- **Education:** Safeguards learning platforms and research environments.

Table 7.6: Impact on Human Order

Aspect	Contribution
Safety	Protects critical infrastructure and human life
Privacy	Ensures secure handling of user data
Social Justice	Enables fair access to digital security
Education	Secures online learning platforms

7.5 Overall Mutual Fulfilment

The proposed system demonstrates a strong alignment between technological advancement and societal well-being, highlighting the concept of mutual fulfilment. It shows that tech-

nological innovation can coexist with environmental sustainability and human values, rather than conflicting with them.

Technological–Environmental Synergy

The system achieves a high detection accuracy of approximately 96% while maintaining minimal computational overhead. This balance indicates that strong cybersecurity measures do not necessarily require excessive resource consumption. By optimizing algorithms and reducing processing time, the system contributes to energy efficiency and environmental sustainability. This synergy between technology and environment supports the development of secure and sustainable digital infrastructures.

Universal Values Integration

The system incorporates fundamental human values within its design and operation. These values are reflected not only in the technical performance but also in the broader impact of the system on society.

- **Truth:** Accurate anomaly detection ensures that threats are identified correctly, with a low false positive rate.
- **Justice:** Provides equal access to cybersecurity solutions across different user groups.
- **Compassion:** Protects vulnerable systems and communities from cyber risks.
- **Peace:** Reduces the likelihood of cyber conflicts and disruptions.
- **Sustainability:** Promotes efficient use of resources and eco-friendly system design.

Table 7.7: Integration of Universal Human Values

Value	System Contribution
Truth	Accurate and reliable anomaly detection
Justice	Equal access to cybersecurity protection
Compassion	Protection of vulnerable systems
Peace	Prevention of cyber conflicts
Sustainability	Efficient and eco-friendly design

Holistic Impact

The multi-layered design of the system reflects the interconnected nature of real-world ecosystems. Just as natural systems maintain balance through multiple interacting layers, the proposed system ensures security by analyzing packet-level, flow-level, and behavioural-level data simultaneously.

This holistic approach demonstrates that cybersecurity can play a vital role in promoting human well-being while preserving environmental balance. By integrating advanced

technology with ethical considerations, the system contributes to a harmonious relationship between digital progress and the natural world.

Overall, the proposed framework highlights that technological innovation, when guided by human values and sustainability principles, can lead to mutual fulfilment across human, environmental, and technological domains.

Chapter 8

Conclusion and Future Scope

8.1 Summary

The project aims to address the critical challenge of detecting Distributed Denial of Service (DDoS) attacks in private cloud environments. With the exponential growth of cloud-based infrastructures and increasing dependency on distributed computing systems, ensuring robust network security has become a fundamental requirement. Cyber threats are becoming more sophisticated, and traditional security mechanisms are no longer sufficient to handle modern attack patterns. This necessitates the development of intelligent, adaptive, and scalable intrusion detection systems.

The primary objective of this project was to design and implement a multi-layer intrusion detection system that leverages machine learning and deep learning techniques to identify anomalous network behavior. The system is structured to analyze network traffic at multiple levels, including packet-level, flow-level, and behavioral-level, thereby providing a comprehensive understanding of network activities.

The packet-level analysis focuses on examining individual packets and identifying anomalies based on structural features such as protocol type, packet size, and TCP flags. The flow-level analysis aggregates packets into flows and evaluates communication patterns, while the behavioral-level analysis captures temporal dependencies and long-term trends in network activity. This layered approach ensures that the system is capable of detecting both simple and complex attacks.

The main outcomes of the project are summarized as follows:

- Development of a robust multi-layer intrusion detection system capable of analyzing network traffic at different levels, thereby enhancing detection capability and system reliability.
- Implementation of a combination of machine learning and deep learning algorithms, including Random Forest, Isolation Forest, Autoencoders, and Gated Recurrent Unit (GRU), to enable both supervised and unsupervised anomaly detection.
- Extensive training and evaluation of the system using benchmark datasets such as CICIDS2017, NSL-KDD, and UNSW-NB15, ensuring that the model is tested against diverse and realistic attack scenarios.
- Achievement of high detection accuracy and improved performance metrics such as precision, recall, and F1-score, indicating the effectiveness of the proposed approach.

- Reduction in false positive rates through the use of multi-layer analysis and ensemble decision-making mechanisms.
- Contribution towards Sustainable Development Goal (SDG) 9: *Industry, Innovation and Infrastructure*, by promoting the development of secure, resilient, and intelligent digital infrastructure.

Overall, the project demonstrates that integrating multiple analytical perspectives significantly enhances the capability of intrusion detection systems, making them more effective in identifying modern cyber threats.

8.2 Conclusions

Based on the experimental results and detailed analysis, several important conclusions can be drawn from this study.

- The proposed multi-layer intrusion detection system provides a significant improvement over traditional single-layer systems. By analyzing network traffic at multiple levels, the system is able to capture both low-level anomalies and high-level behavioral patterns.
- The integration of machine learning and deep learning techniques enhances the detection capability of the system. While machine learning models such as Random Forest provide fast and accurate classification, deep learning models such as GRU enable the detection of time-dependent and evolving anomalies.
- The system demonstrates strong performance across multiple benchmark datasets, including CICIDS2017, NSL-KDD, and UNSW-NB15. This indicates that the model is capable of generalizing well across different types of network environments and attack scenarios.
- The use of ensemble techniques, such as combining outputs from different layers using a voting mechanism, improves the overall robustness and reliability of the system. This approach reduces the chances of misclassification and enhances decision accuracy.
- Behavioural-level analysis plays a crucial role in detecting stealthy and long-term attacks that may not be identified using traditional methods. This highlights the importance of incorporating temporal analysis in intrusion detection systems.
- The system is capable of detecting a wide range of attacks, including DDoS, brute force, botnet, spoofing, and port scanning, demonstrating its versatility and practical applicability.

In conclusion, the proposed system offers a scalable, efficient, and intelligent solution for intrusion detection in private cloud environments. It addresses the limitations of existing systems and provides a strong foundation for future research in the field of cybersecurity.

8.3 Original Contributions

This work makes several original and meaningful contributions to the domain of network security and intrusion detection systems.

- A novel multi-layer intrusion detection framework is proposed, which integrates packet-level, flow-level, and behavioral-level analysis. This layered approach enables comprehensive monitoring and detection of network anomalies.
- The study presents a hybrid detection model that combines multiple machine learning and deep learning techniques, including Random Forest, Isolation Forest, Autoencoder, and GRU. This integration allows the system to leverage the strengths of different algorithms.
- A fusion-based decision-making mechanism is introduced, which combines predictions from different layers to improve overall detection accuracy and reduce false positives.
- The system is evaluated using multiple benchmark datasets, ensuring that the results are reliable, reproducible, and comparable with existing research.
- The research highlights the importance of multi-dimensional analysis in intrusion detection and demonstrates how combining different perspectives leads to better performance.
- The work provides insights into the trade-offs between accuracy, computational complexity, and scalability, which are critical factors in real-world deployment.

These contributions not only improve the current state of intrusion detection systems but also open new avenues for future research and development.

8.4 Limitations

Despite the promising results achieved by the proposed system, there are several limitations that need to be addressed.

One of the major challenges is the high computational cost associated with training deep learning models. Behavioural-level analysis using GRU requires significant processing power and memory, which may not be feasible in resource-constrained environments.

Another limitation is scalability. While the system performs well on benchmark datasets, deploying it in large-scale real-world environments with high network traffic may lead to performance bottlenecks. Efficient load balancing and distributed processing techniques are required to overcome this issue.

The system may also generate false positives during sudden spikes in legitimate network activity, such as large file transfers or peak usage periods. This can affect user trust and requires further optimization of detection thresholds.

Concept drift is another important challenge, where changes in network behavior over time can reduce model accuracy. Since the system relies on historical data for training, it may not adapt quickly to new patterns without retraining.

Additionally, the effectiveness of the system depends on the quality and diversity of the training data. If the dataset does not adequately represent real-world scenarios, the model's performance may be affected.

To address these limitations, adaptive techniques such as incremental learning, sliding window retraining, and drift detection algorithms like ADWIN can be implemented. These methods help maintain system performance in dynamic environments.

8.5 Future Scope

The proposed system provides a strong foundation for further research and development in the field of intrusion detection systems. Several enhancements can be explored in future work.

- **Short-term:** The system can be improved by incorporating advanced machine learning models such as XGBoost, LightGBM, and deep neural networks. Feature engineering and hyperparameter tuning can further enhance detection accuracy.
- **Medium-term:** A real-time intrusion detection system can be developed and deployed on cloud platforms such as AWS, Microsoft Azure, or Google Cloud. Integration with real-time data streams will enable continuous monitoring and faster response to attacks.
- **Long-term:** The system can be scaled to support enterprise-level environments with high traffic volumes. Integration with Security Information and Event Management (SIEM) systems can provide a unified security solution.

- The use of federated learning can be explored to enable collaborative learning across multiple organizations without sharing sensitive data, thereby enhancing privacy and security.
- Edge computing can be integrated to perform anomaly detection closer to the data source, reducing latency and improving response time.
- Explainable AI techniques can be incorporated to improve transparency and help security analysts understand the reasoning behind model predictions.

In addition, future research can focus on developing self-adaptive systems that automatically adjust to changing network conditions. The integration of artificial intelligence with cybersecurity will play a crucial role in building resilient and intelligent defense mechanisms against emerging threats.

Overall, the proposed system lays the groundwork for advanced intrusion detection solutions and has the potential to evolve into a comprehensive cybersecurity framework capable of addressing future challenges.

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Appendix A

Web Page of Publication

Conferences > 2025 3rd International Confer... ?

Multi Layered Anomaly Detection Framework for Network Security using Machine Learning

Publisher: IEEE

[Cite This](#)

 PDF

[Srishti Kaura](#) ; [Vidhi Gupta](#) ; [Soumay Verma](#) ; [Suhani Sahu](#) ; [Ayushi Prakash](#) [All Authors](#)

3
Full
Text Views



Abstract

Document Sections

- I. Introduction
- II. Related Work
- III. System Architecture
- IV. Methodology
- V. Experimental Results

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[Authors](#)

Abstract:

As cyberattacks become increasingly sophisticated, it is not enough to rely on just one way for problem detection. Many traditional systems miss suspicious activities which crosses different parts of a network. In order to address this, we designed a framework that performs network data analysis at three layers: individual packets, flow of data overall, and behaviour of the user. Packet level checks details like headers. Flow level analyze how devices are communicating to identify anomalous traffic. Behaviour level tries to learn normal patterns for users and systems, so it can flag anything out of the ordinary, such as insider threats or advanced attacks: This system combines them for a stronger way of catching tricky threats. This architecture will integrate machine learning models in the backend with a user-friendly dashboard in the front end that provides visibility and management simplified. Test results show that the system is able to detect threats with 96% detection accuracy with false-positive rate under 6%.

Published in: [2025 3rd International Conference on IoT, Communication and Automation Technology \(ICICAT\)](#)

Date of Conference: 05-06 December 2025

DOI: [10.1109/ICICAT68430.2025.11414751](#)

Figure A.1: Web Page of Publication

Appendix B

Publication Certificate



Figure B.1: Publication Certificate

Appendix C

SDG Compliance Certificate

Certificate of Compliance with United Nations Sustainable Development Goals

This is to certify that the project titled **Multi Layered Anomaly Detection Framework for Network Security using Machine Learning** submitted by the following final year students of the Bachelor of Technology in Computer Science and Engineering at Ajay Kumar Garg Engineering College, Ghaziabad has been reviewed and found to be in alignment with the UN Sustainable Development Goals (SDGs) listed below. All efforts have been made to ensure no other SDGs are compromised or negatively impacted.

No.	SDG Name	No.	SDG Name	
1	No Poverty	☐ 10	Reduced Inequalities	☐
2	Zero Hunger	☐ 11	Sustainable Cities & Comm.	✓
3	Good Health & Well-being	✓ 12	Responsible Consumption	☐
4	Quality Education	✓ 13	Climate Action	✓
5	Gender Equality	☐ 14	Life Below Water	☐
6	Clean Water and Sanitation	☐ 15	Life on Land	✓
7	Affordable and Clean Energy	✓ 16	Peace, Justice & Institutions	✓
8	Decent Work & Economic Growth	✓ 17	Partnerships for the Goals	✓
9	Industry, Innovation & Infra.	✓		

Signatures of Students

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Vidhi Gupta

Suhani Sahu

Soumay Verma

Date: May, 2026

Signature of Supervisor

Dr. Ayushi Prakash

Department of Computer Science & Engineering

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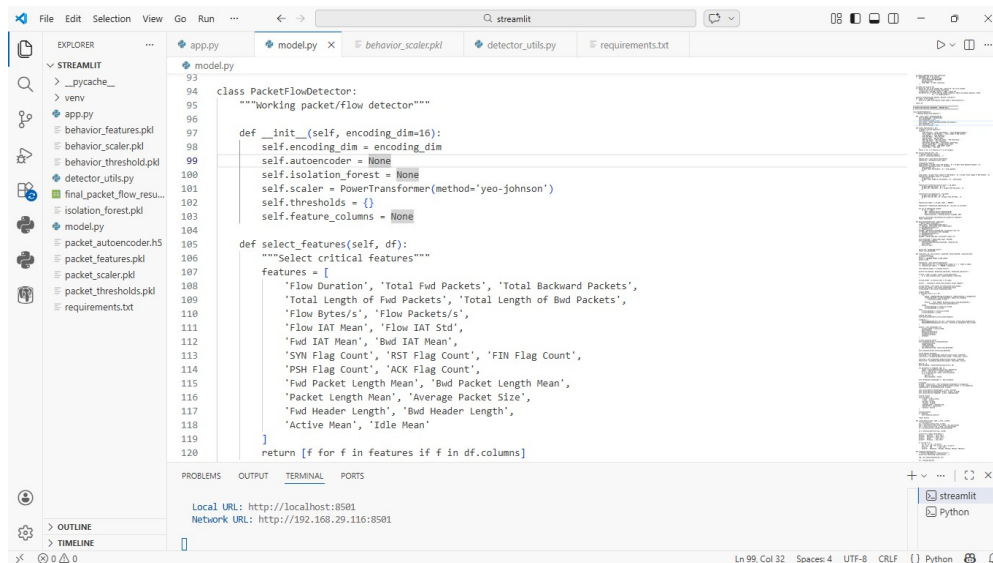
Date: May, 2026

Appendix D

Project Snapshots and Description

D.1 Module 1: [Packet Layer]

Description: The packet level detection module focuses on each network packet individually in an attempt to detect any unusual or abnormal activity. In this case, the Random Forest model is employed in classifying the network packets based on certain features, such as the protocol, ports, packet size, and flags, found in the packet headers. The code snippet demonstrates how the features are being constructed and fed to the trained Random Forest model for prediction. The detection module examines each packet individually, enabling it to detect unusual packets, such as erroneous packets, spoofing, and unusual packet patterns. This module acts as the initial detection system, enabling fast detection of unusual activity in the network packets before moving on to the next level of analysis.



```
93 class PacketFlowDetector:
94     """Working packet/flow detector"""
95
96     def __init__(self, encoding_dim=16):
97         self.encoding_dim = encoding_dim
98         self.autoencoder = None
99         self.isolation_forest = None
100         self.scaler = PowerTransformer(method='yeo-johnson')
101         self.thresholds = {}
102         self.feature_columns = None
103
104     def select_features(self, df):
105         """Select critical features"""
106         features = [
107             'Flow Duration', 'Total Fwd Packets', 'Total Backward Packets',
108             'Total Length of Fwd Packets', 'Total Length of Bud Packets',
109             'Flow Bytes/s', 'Flow Packets/s',
110             'Flow IAT Mean', 'Flow IAT Std',
111             'Fwd IAT Mean', 'Bud IAT Mean',
112             'SVN Flag Count', 'RST Flag Count', 'FIN Flag Count',
113             'PSH Flag Count', 'ACK Flag Count',
114             'Fwd Packet Length Mean', 'Bud Packet Length Mean',
115             'Packet Length Mean', 'Average Packet Size',
116             'Fwd Header Length', 'Bud Header Length',
117             'Active Mean', 'Idle Mean'
118         ]
119         return [f for f in features if f in df.columns]
```

Figure D.1: Screenshot: Module 1 – [Packet layer]

D.2 Module 2: [Flow layer]

Description: The flow level detection module is based on the aggregated level of network traffic, as opposed to individual packets. At this level, the Autoencoder models are trained on what normal behavior looks like in terms of aggregated network flows. The code snippet demonstrates how features such as the duration of the flow, the number of packets and bytes, and the time gap are identified and then passed on to the Autoencoder. The logic behind

using an Autoencoder is that the larger the reconstruction error, the higher the probability of an anomaly. This helps in the detection of attacks such as DDoS, brute force, or botnet activity, as they are more visible when looking at overall communication patterns between devices rather than individual packets.

```

32
33 # =====
34 # PREPROCESSING
35 # =====
36
37 def preprocess_cicids2017_final(df):
38     """final robust preprocessing"""
39     print("% Preprocessing CICIDS2017...")
40
41     df.columns = df.columns.str.strip()
42
43     if 'Label' in df.columns:
44         df['Label'] = df['Label'].str.strip().str.upper()
45
46     numeric_columns = df.select_dtypes(include=[np.number]).columns.tolist()
47
48     # Handle inf/nan
49     for col in numeric_columns:
50         df[col] = df[col].replace([np.inf, -np.inf], np.nan)
51         df[col] = df[col].fillna(df[col].median())
52
53     # Clip outliers
54     for col in numeric_columns:
55         mean = df[col].mean()
56         std = df[col].std()
57         if std > 0:
58             lower = mean - 5 * std
59             upper = mean + 5 * std

```

Figure D.2: Screenshot: Module 2 – [Flow layer]

D.3 Module 3: [Behaviour Layer]

Description: The behavioral level detection module involves the identification of long-term trends with regards to the behavior of the user and the system as a whole. In this case, a GRU neural network is used to identify trends that may be present with regards to the behavior of the network over time. As shown in the code snippet above, the network data is processed and then fed into the GRU network to identify anomalies. By analyzing the trends with regards to the behavior of the network, the system is able to identify sophisticated attacks that may be present, including insider attacks.

This is the highest level of analysis with regards to the network activity. As shown above, this level is able to work in conjunction with the packet and flow levels.

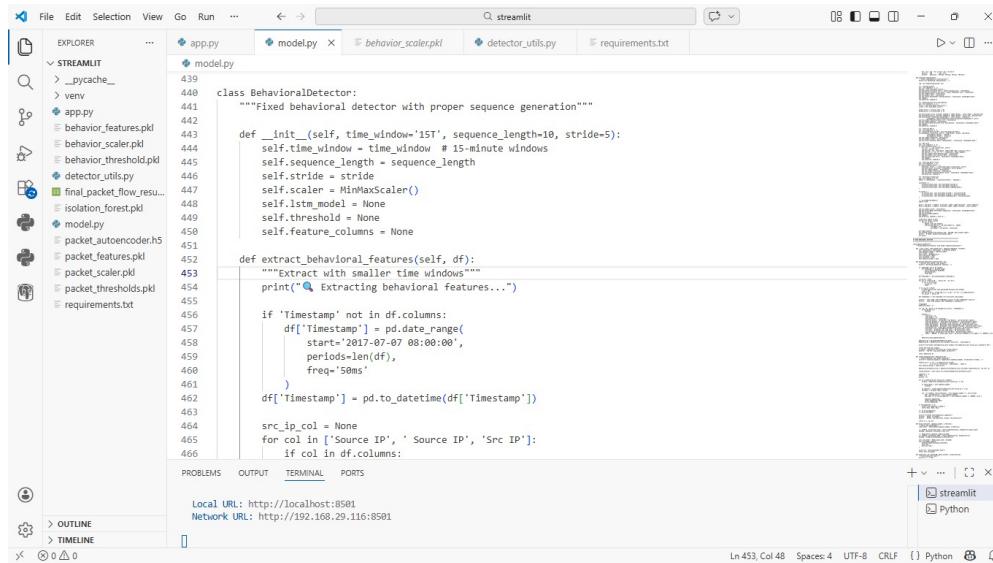


Figure D.3: Screenshot: Module 3 – [Behavioural layer]

D.4 Module 4: [Main Execution]

Description: Fusion Layer

The fusion layer integrates the results obtained from the packet-level, flow-level, and behavior-level detectors to give a conclusive verdict on the network traffic. Each stage individually analyzes the network traffic with its own machine learning model, and the results are combined to give a conclusive verdict on whether the network traffic is normal or malicious. The code snippet shows the integration of results obtained from all three layers to give a conclusive verdict.

The use of multiple layers increases the reliability of the results obtained by reducing false alarms. This gives a holistic view of the network traffic, which increases the ability to detect complex cyber attacks.

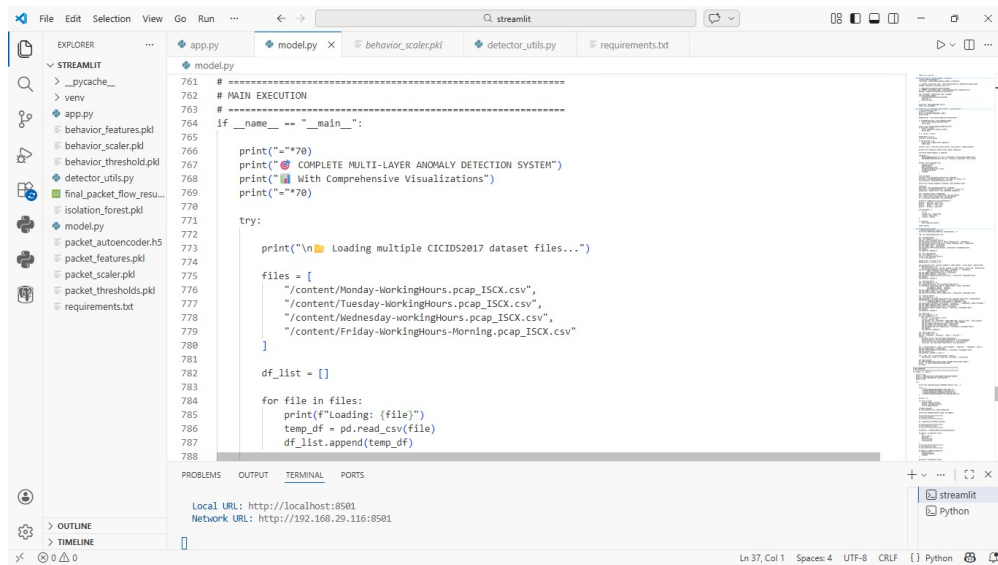


Figure D.4: Screenshot: Module 4 – [Fusion Engine]

Full Source Code Repository:

GitHub Repository Link

Appendix E

Student Curriculum Vitae

Curriculum Vitae

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Passport Size Photo

Educational Qualifications

Degree	Institution	Board/University	Year
B.Tech (CSE)	AKGEC, Ghaziabad	AKTU Lucknow	2026
Class XII	MTMHSS	ICSE	2022
Class X	MTMHSS	ICSE	2020

Technical Skills

Languages	C, C++, Python, Java
Web / UI	HTML, CSS, JavaScript, Flutter WebApp
Databases	MySQL, MongoDB
Tools	Git, VS Code, Linux
Others	Dart, Flutter, Firebase

Projects

1. AI Powered Gym

June 2025 – Aug 2025

Built real-time video streaming with WebSocket integration for instant exercise-form feedback from the ML.

2. Sales Yatra

March 2025 – April 2025

Built a delivery tracking app with Firebase Authentication, real-time delivery status updates, and geofencing alerts.

Internships

GoPool Services Private Limited

June 2025 – Aug 2025

Role: Flutter Developer Intern

Built the AI Gym app with a smooth UI and real-time video streaming using Flutter, Go APIs and WebSockets.

Certifications and Achievements

- Finalist at Devthon Tech Fest (Prometeo), IIT Jodhpur for innovative app-based solution.
 - Conducted workshops on Flutter app development, version control (Git), and mobile UI design, reaching 100+ attendees.
 - Earned certifications in Dart & Flutter including "Complete Flutter Dev Course" and "Full Stack Mobile Dev" from Udemy.
-

Declaration

I hereby declare that all information furnished above is true to the best of my knowledge.

Date: May, 2026

Signature: _____

Curriculum Vitae

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Passport Size Photo

Educational Qualifications

Degree	Institution	Board/University	Year
B.Tech (CSE)	AKGEC, Ghaziabad	AKTU Lucknow	2026
Class XII	CCDPS	CBSE	2022
Class X	CCDPS	CBSE	2020

Technical Skills

Languages	Python, Java, C, HTML/CSS, JavaScript, SQL, Dart
Web / UI	HTML, CSS, JavaScript, Flutter
Databases	MySQL, MongoDB
Tools	Git, VS Code, Linux, Docker
Others	LangChain, Hugging Face Transformers, REST APIs, FAISS, Firebase, NLTK, RAG

Projects

- Multi-Agent-YouTube-ChatBot** 2022–2026
 Developed a multi-agent YouTube chatbot using LangChain, Groq, and Streamlit, capable of transcribing, summarizing, and answering context-aware questions from YouTube videos with memory-enabled conversational flow.
- Malicious Link Analyzer** August

This project is a Phishing URL detection Project . It uses a trained Random Forest Classifier to classify URLs as either Phishing or legit. The model is trained on a dataset containing engineered features that represent characteristics of URLs commonly used in phishing attacks.

Internships / Work Experience

MathonGo

Sept 2025 – Feb 2026

Role: AI Automation Intern

- Designed and deployed a RAG pipeline to match past-year questions from textbooks and online sources, increasing retrieval consistency by 35%.
 - Implemented TF-IDF + semantic search and automated web scraping workflows, scaling question dataset coverage by 2×.
 - Integrated the Gemini API to auto-generate explanations and hints, reducing manual content creation time.
-

Certifications and Achievements

- Deloitte Australia Data Analytics Job Simulation on Forage – (June 2025)
 - Secured National Semi-Finalist position in Flipkart GRID 7.0 among 160,000+ participants.
 - Qualified for Regional Level of the Google Developer Student Clubs Solution Challenge 2024 – led a team to build a tech solution aligned with UN Sustainable Development Goals
-

Declaration

I hereby declare that all information furnished above is true to the best of my knowledge.

Date: May, 2026

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Curriculum Vitae

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Passport Size Photo

Educational Qualifications

Degree	Institution	Board/University	Year
B.Tech (CSE)	AKGEC, Ghaziabad	AKTU Lucknow	2026
Class XII	SSP	CBSE	2021
Class X	SSP	CBSE	2019

Technical Skills

Languages	C, C++, Python, Java
Web / UI	Streamlit FastAPI REST APIs
Databases	SQL Databases FAISS ChromaDB
Tools	TensorFlow PyTorch Hugging Face Transformers OpenCV YOLO
Others	Generative AI & LLMs (GPT/OpenAI APIs, LangChain, LangGraph, RAG, Prompt Engineering, ReAct Agents)

Projects

1. AI Mental Health Therapist

Jan 2026–Feb 2026

Built an LLM-powered conversational agent using LangGraph and ReAct agents that provides empathetic responses and mental health guidance through tool calling and contextual

reasoning.

2. AI Data Validation System

Oct 2025–Dec 2025

Developed an agentic AI pipeline that automatically validates, cleans, and detects anomalies in datasets using machine learning and rule-based validation techniques.

Internships

Smaran AI

Aug 2025 – Dec 2025

Role: AIML Intern

Developed and deployed an AI pipeline for PDF text and image extraction using OCR and NLP, processing 500+ documents with 92% OCR accuracy to enable multimodal question answering.

Certifications and Achievements

- Finetuning Large Language Models — DeepLearning.AI (LLM Fine-tuning, Prompt Engineering, Model Adaptation)
 - Introduction to Model Context Protocol (MCP) — Anthropic (MCP Architecture, MCP Servers & Clients, Tool Integration, Document Management System)
 - 100 Days of Code™: The Complete Python Pro Bootcamp — Udemy (Python Programming, Automation, Web Development, Data Handling)
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Declaration

I hereby declare that all information furnished above is true to the best of my knowledge.

Date: May, 2026

Signature: _____

Curriculum Vitae

Full Name Suhani Sahu
Roll No. 2200271540113
Branch Computer Science & Engineering
Batch 2022 – 2026
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LinkedIn linkedin.com/in/sahusuhani
GitHub github.com/shutupsuhani



Passport Size Photo

Educational Qualifications

Degree	Institution	Year
B.Tech (CSE)	AKGEC, Ghaziabad	2026
Class XII	SVSVM, Ghaziabad	2022
Class X	SVSVM, Ghaziabad	2020

Technical Skills

Languages	C++, Java, Python, JavaScript, TypeScript
Frontend	HTML, CSS, React.js, Next.js
Backend	Node.js, Express.js, REST APIs
Databases	MongoDB, MySQL
Tools	Git, VS Code, Postman

Projects

Voice Notes App	2025
AI-based voice recording, transcription and summarization system.	
Quick-Prep AI	2025
AI-powered interview preparation platform built using Next.js.	

Internships

Newgen Software, Noida	Feb 2026 – Present
Software Engineer Intern (Java, MySQL).	
Abhi Loans	Sep 2025 – Dec 2025
Software Engineer Intern (React.js, TypeScript).	

Achievements

- Meesho Women in Tech Challenge – IdeaFunnel Round.
- PwC Launchpad 3.0 – Top 2 at AKGEC.
- Solved 300+ problems on LeetCode.

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Date: May, 2026

Signature: _____